

STANDARDS AND INFORMATION DOCUMENTS

AES60-2011



AES standard for audio metadata— Core audio metadata

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AES standard for audio metadata - Core audio metadata

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Abstract

This specification addresses the creation, management and preservation of material that can be re-used as originally produced, or may provide input material for new production projects. Material is expected to be exchanged between various organisations or between production facilities in a distributed environment.

The core set of metadata presented in this specification is a co-publication of EBU Tech3293-2008 EBU Core, itself an extension to and a refinement of the Dublin Core. EBUCore is a minimum list of attributes characterizing video and / or audio media resources. An XML representation is also provided in case this metadata would be implemented, for example in archive exchange projects using the Open Archive Initiative's Protocol for Metadata Harvesting (OAI-PMH).

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Foreword

This foreword is not part of the AES60-2011 *AES standard for audio metadata - Core audio metadata*.

This document has its roots in the AES project X098A where the concept of providing a core administrative metadata set for audio exchange explored. This metadata set was based on Dublin Core and expanded to meet the basic requirements of the audio industry. It was well ahead of its time and as a result did not attract sufficient industry traction to complete until comparatively recently where similar work covering both audio and video was carried out within the EBU. The decision was eventually taken to base the AES core descriptive metadata on the EBU core and co-publish this under the auspices of EBU and AES, which is expected to have a greater impact on the industry as a whole.

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Note on normative language

In AES standards documents, sentences containing the word “shall” are requirements for compliance with the document. Sentences containing the verb “should” are strong suggestions (recommendations). Sentences giving permission use the verb “may”. Sentences expressing a possibility use the verb “can”.



AES standard for audio metadata - Core audio metadata

Introduction

The “AES Core” set of metadata defined in this specification has been identified as being the minimum information needed to describe audio media resources.

Modern IT-based production environments need metadata to identify and retrieve content correctly and efficiently. Metadata provides an important link between various production operations and the following core set of information is a minimum requirement for practical operations. Content archives provide a common basis to describe content in a way that supports other processes in the production chain.

The decision to issue AES Core as a co-publication with the EBU Core is seen as an important factor in maintaining interoperability between the audio and video industries. For this reason, this specification includes elements that may seldom be used in a purely audio context. However, much audio material is destined to coexist with video and other media at some point during its life, so these additional elements may not be superfluous. It is not expected that the AES will actively develop the non-audio aspects of this standard but rather rely on active cooperation with EBU to co-maintain the AES and EBU cores.

This “AES Core” metadata set is primarily a minimum list of elements and attributes for which an XML representation is also proposed. Being based on the EBU Core, it follows that it is also an extension to the Dublin Core metadata set. The Dublin Core is being used as a core metadata set by librarians and in cultural heritage projects. The AES Core and EBUCore, used for radio and television archives, offer a bridge between cultural-heritage databases, broadcasting production systems and broadcasting archive repositories.

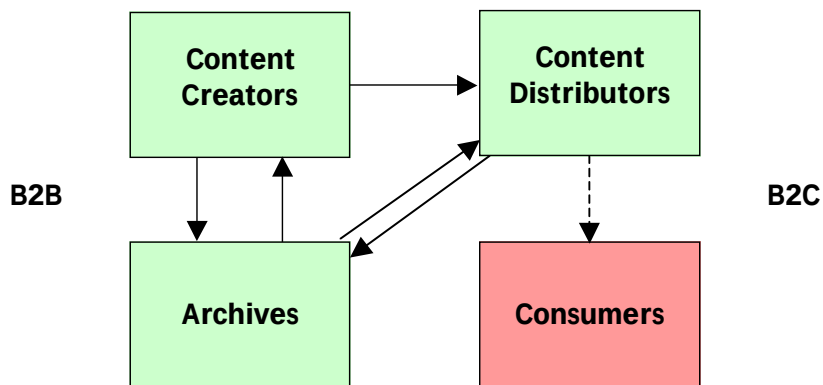


Figure 1: Archiving - a core process to define metadata

0 Preamble

0.1 Documentation conventions

A mono-spaced typeface is used to identify computer code expressions to distinguish them from regular text. Examples are shown using XML structural conventions.

1 Scope

This specification addresses the creation, management and preservation of material that can be re-used as originally produced, or may provide input material for new production projects. Material is expected to be exchanged between various organisations or between production facilities in a distributed environment.

The core set of metadata presented in this specification is based on the EBUCore, itself an extension to the Dublin Core. It is a minimum list of attributes characterizing audio and / or video media resources. An XML representation is also proposed in case this metadata would be implemented, for example in archive exchange projects using the Open Archive Initiative's Protocol for Metadata Harvesting (OAI-PMH).

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 15836, *Dublin Core Metadata Element Set*, International Standards Organisation, Geneva, Switzerland.

XML schema, World Wide Web Consortium, www.w3.org

Namespaces in XML, World Wide Web Consortium, www.w3.org

RFC 2045 *Multipurpose Internet Mail Extensions (MIME) Part One: Format of Internet Message Bodies*, The Internet Society, www.isoc.org

RFC 3339 *Date and Time on the Internet: Timestamps*, The Internet Society, www.isoc.org

3 Definitions and abbreviations

3.1

Unique Resource Identifier

URI

<http://tools.ietf.org/html/rfc3986>

3.2

Unique Resource Namespace

URN

<http://tools.ietf.org/html/rfc3986>

3.3

Unique Resource Locator

URL

<http://tools.ietf.org/html/rfc3986>



4 Core Metadata Set

4.1 General

4.1.1 Precedence

The AES Core is based on the EBUCore metadata specification. It is a collection of basic metadata elements to be used to describe content in metadata environments based on the Dublin Core Metadata Element Set.

In case of conflict between any detail of this standard and the current version of the associated published schema, the schema shall take precedence.

4.1.2 Element conventions

The characterization and semantics of each element shall correspond to the information as outlined below:

n.1.1.1 This is the name of the element

Cardinality	this specifies the number of times an elements can be used when describing a piece of content
Requirement	this states whether is element is required or optional
Definition	this provides a short description of the element and its scope of use
Format	defines the type or format of the element. For example: text or URI
Schema	give a syntactic view of the element representation in the AES-Core schema
Example	this gives an informative illustration of the element
Refinement	this is an informative list of elements and attributes that may be used to further define the element in a structured way

NOTE 1 In this document the name is preceded by a sub-clause number to aid human readability. Subclause numbering is not part of the metadata set and may change in later versions of this document.

NOTE 2 Mapping may exist between this standard and equivalent elements in other standards.

4.1.3 The XML instance of EBU Core and AES Core

The standard defines an XLM-schema, and so the data instances of the standard will be XML-documents. The structure is defined in the schema but will typical be something like:

```
<?xml version="1.0" encoding="UTF-8"?>
<ebuCoreMain xmlns="urn:ebu:metadata-schema:ebuCore_2010"
xmlns:dc="http://purl.org/dc/elements/1.1/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="urn:ebu:metadata-schema:ebuCore_2010
http://www.aes.org/standards/schemas/aes60-2011.xsd">
  <coreMetadata>
    <title>
      <dc:title>Some title</dc:title>
    </title>
  </coreMetadata>
</ebuCoreMain>
```

The element name **ebuCoreMain** is common for EBU Core and AES60 to ensure that different XML instances are compatible.



4.1.4 Implementation guidelines

4.1.4.1 General remarks

Several aspects of the specification are left to the implementer (for example, regarding the mapping to pre-existing in-house metadata schemas).

Reference data identified in 4.2 is proposed by default but can be extended or replaced. In order to maximise interoperability in case of exchange, for example, it is recommended that extensions or alternative reference data be duly documented, maintained and made available to other users - as open resources on the Internet, for example.

The schema proposed in Annex F is built as an extension to the Simple Dublin Core and further developed by the EBU to facilitate transformation to the Simple Dublin Core representation as required by certain applications such as the European Digital Library. For the same reason, it is recommended that predefined 'relations' be used when available.

4.1.4.2 Use of the attribute groups

4.1.4.2.1 typeGroup, formatGroup, statusGroup

Each of these groups consists of the same three attributes: a 'label', a 'definition' and a 'link'.

label	gives the actual name given to the <i>type</i> , <i>format</i> or <i>status</i> . For example, 'episode' will be used to specify that the type of Title provided is the title of an episode.
definition	can be used to provide either a definition of the <i>type</i> , <i>format</i> or <i>status</i> , or the name of the reference data set where this <i>type</i> , <i>format</i> or <i>status</i> is defined.
link	is used to provide a URI to a Classification Scheme (in complement to the 'label') or to a Classification Scheme term (in this case the use of the label is optional).

4.1.4.2.2 dateGroup

This group is associated with only a title or alternative title. It consists of three attributes: 'startDate', 'endDate', and 'period'.

Precise date and/or time periods shall be defined using the 'startDate' and 'endDate'.

The 'period' attribute is used to define a period by its name, for example 'April 2000'.

4.1.4.3 External reference data

Lists of controlled terms are handled by external Classification Schemes structured to allow access to terms from a predefined hierarchical vocabulary list (thesaurus). Each list is uniquely identified by its namespace (URI in the form of a URN or URL) and 'Alias'. A Classification Term is defined by a unique key (**termID**) or a name as follows:

EXAMPLE:

```
<ClassificationScheme uri="urn:ebu:metadata-cs:ContentGenreCS:2008">
  <Alias>GenreCS</Alias>
  <Term termID="3.1">
    <Name xml:lang="en">NON-FICTION / INFORMATION</mpeg7:Name>
    <Term key="3.1.1">
      <Name xml:lang="en">News</mpeg7:Name>
    </Term>
    <!--etc.-->
  </ClassificationScheme>
```

It is an important implementation requirement to ensure that these resources are accessible by the metadata recipient. Classification schemes shall preferably be available as resources on the open Internet via maintained URLs. In this case URIs shall respect the following syntax:

URL#termID

EXAMPLE: http://www.ebu.ch/metadata/cs/ebu_ContentGenreCS.xml#3.1

A conforming parser uses that URI to resolve the **termID** reference to a resource, whether physical or logical. Once the **termID** has been resolved, the term name can be accessed (for example: 'News' in the above example). This resolution method is left to the discretion of each recipient.

URIs (URLs) can be replaced by aliases to provide a more concise, application-specific way of referring to classification terms as long as a look-up table is provided describing the relationship between aliases and URIs.

If '**GenreCS**' is the alias for "http://www.ebu.ch/metadata/cs/ebu_ContentGenreCS.xml", in the above example 'News' will be accessed through "**GenreCS#3.1**".

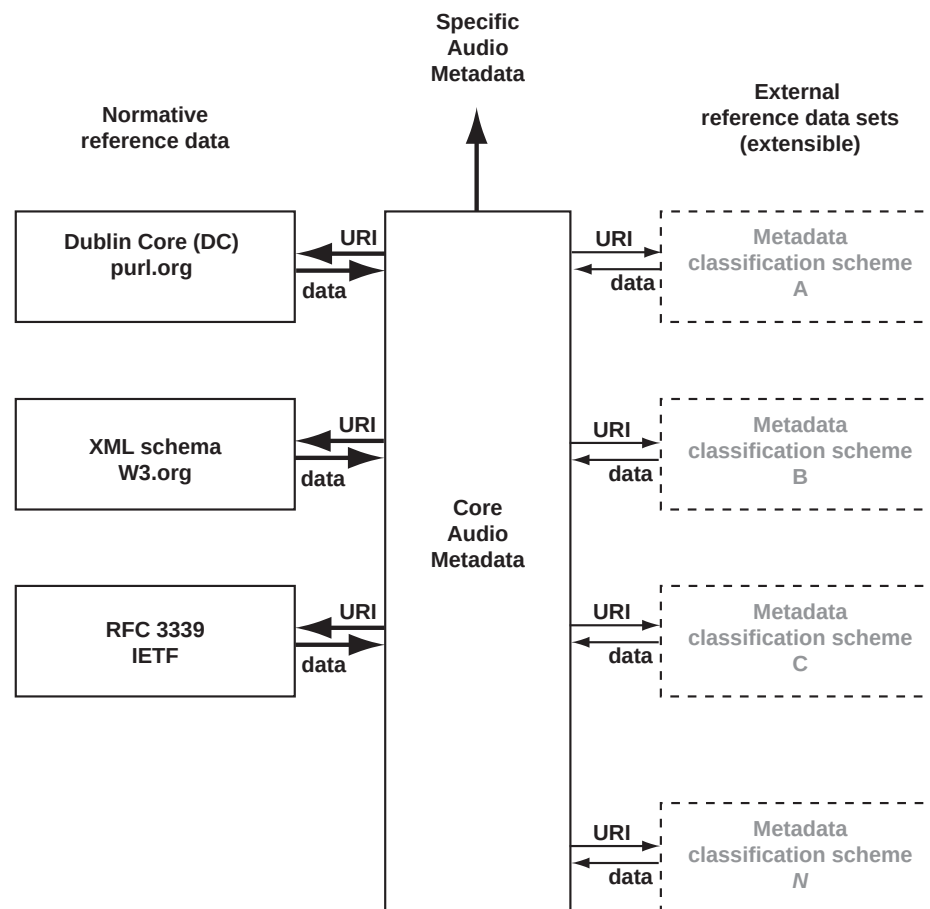


Figure 2 - Relationship between core metadata and external metadata sets

4.2 Core metadata-set elements and semantics

4.2.1 Title

4.2.1.1 Title

Cardinality	Multiple occurrences of the same Title are possible in different languages
Requirement	Mandatory
Definition	A Title is the ‘main’ name given to a resource, for example a media item, a media object, or a sequence as specified by the associated title type. It corresponds for a series to the series title, for a programme to the programme title, for an item to the item title, etc. Titles are recorded as they appear. The Title is the name by which a resource is formally known and that everyone should use to refer to or search for that particular resource. The Title may be provided in several languages, described by the attribute xml:lang It is sometimes common practice to put dates into the Title. If present, the startDate (indicating when the Title was attributed) in the date attribute should be consistent.
Format	Free text
Schema	/coreMetadataType/title/dc:title
Example	<pre><coreMetadata> <title> <dc:title>Twilight zone</dc:title> </title> <title> <dc:title xml:lang="fr">La quatrième dimension</dc:title> </title> </coreMetadata></pre>
Refinement	Title Date, Title Note (optional additional contextual information)

4.2.1.2 Alternative Title

Cardinality	Multiple
Requirement	Optional
Definition	An Alternative Title is the name other than the ‘main’ Title given to a resource, for example: a media item, a media object, or a sequence. Alternative Titles are recorded as they appear. An Alternative Title may be attributed to a resource for several reasons described using the status attributes.
Format	Free text
Schema	/coreMetadataType/alternativeTitle/dc:title
Example	<pre><alternativeTitle> <dc:title>The time machine lost in space</dc:title> </alternativeTitle></pre>
Refinement	Title Status, Title Date, Title Note (optional additional contextual information)

4.2.1.2.1 Alternative Title Status

Cardinality	Unique per Alternative Title
Requirement	Optional
Definition	The Title Status descriptor indicates the sort of Title or Alternative Title being defined. The status 'main' is reserved for the Title and should not be used for Alternative Titles.
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	statusGroup of attributes: /coreMetadataType/alternativeTitle/@statusLabel /coreMetadataType/alternativeTitle/@statusLink and (optional) /coreMetadataType/alternativeTitle/@statusDefinition
Reference data	ebu_TitleStatusCodeCS
Example	<pre><alternativeTitle statusLabel="working" statusLink="http://www.ebu.ch/metadata/cs/ebu_TitleStatusCodeCS.xml#6" statusDefinition="a temporary title, which is different from the formal title under which the content has been published"> <dc:title>The time machine lost in space</dc:title> </alternativeTitle></pre>

4.2.1.2.2 Title Date / Alternative Title Date

Cardinality	Unique per Title or Title Alternative
Requirement	Optional
Definition	Defines the date of creation/attribution, or the associated period for the Title or Alternative Title
Format	Date Group (see 4.2.7.6)
Schema	dateGroup of attributes: /coreMetadataType/title/@startDate and /coreMetadataType/title/@endDate or /coreMetadataType/title/@period
Example	<pre><alternativeTitle startDate="2010-10-10" startTime="09:00:00+01:00" endDate="2010-11-11" endTime="19:00:00+01:00"> <dc:title>The time machine lost in space</dc:title> </alternativeTitle></pre>

4.2.2 Creator

Cardinality	Multiple
Requirement	Optional
Definition	The descriptor creator identifies an ‘entity’ (a person, group of persons or organisation) primarily responsible for creating the content of the resource - <u>behind the camera</u> . Different roles may be considered as representing a creator, for example: a producer, an author, etc.
Format	Entity Type (see 4.2.19)
Schema	/coreMetadataType/creator
Reference data	ebu_RoleCodeCS
Example	<pre> <creator> <contactDetails> <name> <givenName>Wolfgang Amadeus</givenName> <familyName>Mozart</familyName> </name> <address> <addressLine>Irgentwogasse 133</addressLine> <addressTownCity>Wien</addressTownCity> <country>Österreich</country> </address> <contactOccupation>Entertainer</contactOccupation> </contactDetails> <role typeLabel="Composer" typeLink="http://www.ebu.ch/metadata/cs/ebu_RoleCodeCS.xml#17.1.7"/> </creator> </pre>

4.2.3 Subject

Cardinality	Multiple
Requirement	Optional
Definition	The generalised topic that represents the intellectual content of the resource. Typically, a subject is expressed by keywords, key phrases. Controlled vocabularies, authorities, or formal classification schemes (codes) may be employed when selecting descriptive subject terms. Persons as subjects are also placed here. Genre of the content is placed under element “Type”.
Format	Classification Scheme (code) or free text
Schema	/coreMetadataType/subject/dc:subject and/or /coreMetadataType/subject/subjectCode and (optional) /coreMetadataType/subject/subjectDefinition
Reference data	Library of Congress Subject Heading (LCSH) Library of Congress Classification (LCC) Medical Subject Headings (MeSH) Dewey Decimal Classification (DDC) Dansk decimalklassedeling 5.utgave (DK5) Klassifikasjonssystem för svenska bibliotek (SAB)



Universal Decimal Classification (UDC)
 Norske emneord
 ebu_IPTCSubjectCodeCS
 etc.

Example	<pre> <subject> <dc:subject>Tennis</dc:subject> <subjectCode>15065000</subjectCode> <subjectDefinition>the subject is about tennis (sport, game)</subjectDefinition> </subject> </pre>
Refinement	Subject Type, Note (additional contextual information)

4.2.3.1 Subject Type

Cardinality	Unique per Subject
Requirement	Optional
Definition	To define the subject reference in use such as the reference data source
Format	Classification scheme link (reference data file URL and an optional classification scheme term code) or free text (label) and an optional definition
Schema	typeGroup of attributes: /coreMetadataType/subject/@typeLabel and/or /coreMetadataType/subject/@typeLink and (optional) /coreMetadataType/subject/@typeDefinition
Reference data	Library of Congress Subject Heading (LCSH) Library of Congress Classification (LCC) Medical Subject Headings (MeSH) Dewey Decimal Classification (DDC) Dansk decimalklassedeling 5.utgave (DK5) Klassifikasjonssystem för svenska bibliotek (SAB) Universal Decimal Classification (UDC) Norske emneord
Example	<pre> <subject typeDefinition="http://www.ebu.ch/metadata/cs/ebu_IPTCSubjectCodeCS.xml" typeLabel="http://www.ebu.ch/metadata/cs/ebu_IPTCSubjectCodeCS.xml" typeLink="http://www.ebu.ch/metadata/cs/ebu_IPTCSubjectCodeCS.xml"> <dc:subject>Tennis</dc:subject> <subjectCode>http://www.ebu.ch/metadata/cs/ebu_IPTCSubjectCodeCS.xml#1 5065000</subjectCode> <subjectDefinition>the subject is about tennis (sport, game)</subjectDefinition> </subject> </pre>

4.2.4 Description

Cardinality	Multiple
Requirement	Optional
Definition	Free-form text or a narrative to report general notes, abstracts, or summaries about the intellectual content of a resource. The information may be in the form of a paragraph giving an individual program description, anecdotal interpretations, or brief content reviews. The description may also consist of outlines, lists, bullet points, edit decision lists, indexes, or tables of content, a reference to a graphical representation of content or even a pointer (URI, URL) to an external resource. A running order can also be provided as a description. For a Radio or television programme a running order can be used as description. A description can be provided in different languages.
Format	Free text
Schema	<code>/coreMetadataType/description/dc:description</code>
Example	<pre><description typeLabel="Abstract"> <dc:description>A news report about an important international event focusing on the following list of important highlights.</dc:description> </description></pre>
Refinement	Description Type, Description Note (optional additional contextual information)

4.2.4.1 Description Type

Cardinality	Unique per Description
Requirement	Optional
Definition	To define the form of presentation for the information.
Format	Classification scheme link (reference data file URL and an optional classification scheme term code) or free text (label) and an optional definition
Schema	typeGroup of attributes: <pre>/coreMetadataType/description/@typeLabel and/or /coreMetadataType/description/@typeLink and (optional) /coreMetadataType/description/@typeDefinition</pre>
Reference data	ebu_DescriptionTypeCodeCS
Example	<pre><description typeLabel="Promotional Information" typeLink="http://www.ebu.ch/metadata/cs/ebu_DescriptionTypeCodeCS.xml#5 "> <dc:description>A news report ... highlights.</dc:description> </description></pre>

4.2.5 Publisher

Cardinality	Multiple
Requirement	Optional
Definition	A publisher is a person, an organization, or a service. Typically, the name of a Publisher should be used to indicate the entity primarily responsible for distributing or making a resource available to others; for example by broadcasting, selling, leasing, renting and other modes of distribution.
Format	Entity Type (see 4.2.19)
Schema	<code>/coreMetadataType/publisher</code>
Example	<pre> <publisher> <contactDetails> <name> <name>BBC</name> </name> </contactDetails> <role typeLabel="Distributor" typeLink="http://www.ebu.ch/metadata/cs/ebu_RoleCodeCS.xml#10.2"/> </publisher> </pre>

4.2.6 Contributor

Cardinality	Multiple
Requirement	Optional
Definition	The descriptor contributor identifies a person or organization that has made substantial creative contributions to the content of a resource. Refers particularly (but not only) to participation <u>in front of the camera</u>. If in doubt whether an entity is a creator or contributor use the element contributor.
Format	Entity Type (see 4.2.19)
Schema	<code>/coreMetadataType/contributor</code>
Example	<pre> <contributor> <contactDetails> <name> <givenName>Andy</givenName> <familyName>Williams</familyName> </name> </contactDetails> <role typeLabel="Presenter" typeLink="http://www.ebu.ch/metadata/cs/ebu_RoleCodeCS.xml#25.10"/> </contributor> </pre>

4.2.7 Date

Cardinality	Multiple
Requirement	Optional
Definition	Dates associated with events occurring during the life of the resource. Typically, Date will be associated with the creation, modification or availability of the resource.
Format	Any string describing a date, for a more strict date use the refinement elements
Schema	<code>/coreMetadataType/date/dc:date</code>
Refinement	Date Issued, Date Created, Date Modified, Date Digitised, Date Alternative
Example	<code><!--All dates can have the same start and end attributes, alternative</code>



```

date also a type-->
<date>
  <created startYear="1789" endYear="1791"/>
  <issued startDate="1967-04-05"/>
  <modified startDate="1968-04-21" startTime="09:35:00+01:00"/>
  <digitised startDate="2001-09-10" startTime="21:34:57"/>
  <alternative typeLabel="Date Ingested " typeDefinition="date at which
content was fed into the production system" startDate="2010-03-27"
startTime="10:34:47"/>
  <alternative typeLabel="Date published" typeDefinition="timespan for
publishing of the content on the web" startDate="2010-01-01"
startTime="00:00:00Z" endDate="2010-06-30" endTime="24:00:00"/>
</date>

```

4.2.7.1 Date Issued

Cardinality Unique

Requirement Optional

Definition Date of formal issuance (for example, publication) of the resource.

Specifies the formal date for a particular version or rendition of a resource has been made ready or officially released for distribution, publication or consumption, for example the broadcasting date of a radio programme.

A specific time may also be associated with the date.

Format Date Group (see 4.2.7.6)

Schema /coreMetadataType/date/issued

4.2.7.2 Date Created

Cardinality Unique

Requirement Optional

Definition To specify the creation date for a particular version or rendition of a resource across its life cycle. It is the moment in time that the media item was finalized during its production process and is forwarded to other divisions or agencies to make it ready for publication or distribution. A specific time may also be associated with the date.

Format Date Group (see 4.2.7.6)

Schema /coreMetadataType/date/created

4.2.7.3 Date Modified

Cardinality Unique

Requirement Optional

Definition Date on which the resource was last changed

Format Date Group (see 4.2.7.6)

Schema /coreMetadataType/date/modified

4.2.7.4 Date Digitised

Cardinality	Unique
Requirement	Optional
Definition	Date on which the resource was digitised.
Format	Date Group (see 4.2.7.6)
Schema	/coreMetadataType/date/digitised

4.2.7.5 Date Alternative

Cardinality	Multiple
Requirement	Optional
Definition	An alternative particular date for which the type can be defined
Format	Date Group (see 4.2.7.6)
Schema	/coreMetadataType/date/alternative
Refinement	Date AlternativeType

4.2.7.5.1 Date Alternative Type

Cardinality	Unique per Alternative
Requirement	Optional
Definition	To define a type of date other than the pre-defined dates (created, etc.).
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	typeGroup of attributes: /coreMetadataType/date/alternative/@typeLabel or /coreMetadataType/date/alternative/@typeLink and (optional) /coreMetadataType/date/alternative/@typeDefinition

4.2.7.6 Date Group

Cardinality	One per date refinement element
Requirement	Optional
Definition	A set of attributes defining a date
Format	@startYear – qYear - To express a start year @startDate – date - To express a start date @startTime – time – To express a start time @endYear – qYear - To express an end year @endDate – date - To express an end date @endTime – time – To express an end time The format types are the types defined in XML Schema
Schema	Applies to several elements in the schema



4.2.8 Type

Cardinality	Multiple
Requirement	Optional
Definition	The nature or genre of the resource. Type includes terms describing general categories, functions, genres, or aggregation levels for content. Recommended best practice is to select a value from a controlled vocabulary. To describe the physical or digital manifestation of the resource, use the Format element (see 4.2.9).
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (dc:type) and an optional definition
Schema	typeGroup of attributes: /coreMetadataType/type/dc:type and/or /coreMetadataType/type/genre/ and (optional) /coreMetadataType/type/objectType/
Reference data	<i>For genre:</i> ebu_ContentAlertSchemeCodeCS ebu_ContentGenreCS ebu_EditorialFormatCodeCS ebu_IntentionCodeCS tva_ContentCommercialCS tva_ContentAlertCS ebu_IntendedAudienceCodeCS <i>For objectType:</i> ebu_ObjectTypeCS
Example	<pre><type> <objectType typeLabel="programme"/> </type> <type> <genre typeLabel="News" typeDefinition="The material is about information and reporting such as daily news"/> </type></pre>
Refinement	Type @Type Group, Type @Note (optional additional contextual information)

4.2.9 Format

Cardinality	Unique per manifestation of a resource
Requirement	Optional
Definition	The physical or digital manifestation of the resource. Use the descriptor Format to identify the format of a particular resource as it exists in its physical or digital form. ‘Physical form’ means an actual physical form that occupies physical space, for example a tape. ‘Digital form’ means a digital file residing on a server or hard drive. Format may be used to determine the software, hardware or other equipment needed to display



or operate the resource.

Format	Free text
Schema	<code>/coreMetadataType/format/dc:format</code>
Example	<pre><format> <dc:format>MPEG video</dc:format> </format></pre>
Refinement	Format Width, Format Height, Format Medium, Format Mime Type, Format Video Format, Format Audio Format, Format File Format, Format Captioning Format, Format Signing Format, Format Start, Format Duration, Format File Size, Format Filename, Format Locator, Format Technical Attribute String, Format Technical Attribute Integer, Format Technical Attribute Flag, Format Technical Attribute Float

4.2.9.1 Format Width

Cardinality	Unique per Format.
Requirement	Optional
Definition	The width of the image or picture. Used as numerator to define the aspect ratio for video content.
Format	Non-negative Integer
Schema	<code>/coreMetadataType/format/width</code>
Example	<pre><format> <width unit="pixel">1920</width> </format></pre>
Refinement	Unit

4.2.9.2 Format Height

Cardinality	Unique per Format.
Requirement	Optional
Definition	The height of the image or picture. Used as denominator to define the aspect ratio for video content.
Format	Non Negative Integer
Schema	<code>/coreMetadataType/format/height</code>
Example	<pre><format> <height>300</height> </format></pre>
Refinement	Unit

4.2.9.3 Format Medium

Cardinality	Multiple
Requirement	Optional
Definition	The material or physical carrier of the resource. If a file, it should be the carrier format.
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	typeGroup of attributes:



	/coreMetadataType/format/medium/@typeLabel
	and/or
	/coreMetadataType/format/medium/@typeLink
	and (optional)
	/coreMetadataType/format/medium/@typeDefinition
Reference data	ebu_StorageMediaTypeCode (extension to IBTN, EBU Tech Doc 3279)
Example	<pre> <format> <medium typeLabel="D5 format HDTV digital television tape" typeLink="http://www.ebu.ch/metadata/cs/ebu_StorageMediaTypeCodeCS.xml# D5H"/> </format> </pre>

4.2.9.4 Format MIME Type

Cardinality	Unique per Format
Requirement	Optional
Definition	The main MIME type of the resource as registered with IANA (see RFC 2045).
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	typeGroup of attributes: /coreMetadataType/format/mimeType/@typeLabel or /coreMetadataType/format/mimeType/@typeLink and (optional) /coreMetadataType/format/mimeType/@typeDefinition
Reference data	MIME Type (http://www.iana.org/assignments/media-types/) ebu_MediaTypeCS

4.2.9.5 Format Video Format

Cardinality	Unique per Encoding
Requirement	Optional
Definition	To provide information on the Video Format in complement to video encoding information on (for example) colour, greyscale or black and white colour schemes, frame rate, sampling rate, scanning format.
Format	Sets of elements describing the video
Refinement	Aspect Ratio, Video Encoding, Video Track, Video Technical Attribute String, Video Technical Attribute Integer, Video Technical Attribute Flag, Video Technical Attribute Float

4.2.9.5.1 Aspect Ratio

Cardinality	Unique per Video Format
Requirement	Optional



Definition	The aspect relation of the video
Format	String with format definition group, additional free text note
Schema	formatGroup of attributes, and note: /coreMetadataType/format/videoFormat/aspectRatio/@formatLabel or /coreMetadataType/format/videoFormat/aspectRatio/@formatLink and (optional) /coreMetadataType/format/videoFormat/aspectRatio/@formatDefinition (optional) /coreMetadataType/format/videoFormat/aspectRatio/@note

4.2.9.5.2 Video Encoding

Cardinality	Unique per Video Format
Requirement	Optional
Definition	The format of the video encoding used
Format	Type group set of attributes
Schema	typeGroup of attributes: /coreMetadataType/format/videoFormat/videoEncoding/@typeLabel or /coreMetadataType/format/videoFormat/videoEncoding/@typeLink and (optional) /coreMetadataType/format/videoFormat/videoEncoding/@typeDefinition

4.2.9.5.3 Video Track

Cardinality	Multiple per Video Format
Requirement	Optional
Definition	The format of the video track or tracks
Format	Type group set of attributes, with additional Video track id and name
Schema	typeGroup of attributes: /coreMetadataType/format/videoFormat/videoTrack/@typeLabel or /coreMetadataType/format/videoFormat/videoTrack/@typeLink and (optional) /coreMetadataType/format/videoFormat/videoTrack/@typeDefinition (optional) /coreMetadataType/format/videoFormat/videoTrack/videoTrackId (optional) /coreMetadataType/format/videoFormat/videoTrack/videoTrackName

4.2.9.5.4 Video Technical Attribute String

Cardinality	Multiple per Video Format
Requirement	Optional
Definition	Additional properties of the video resource



Format String with type and format group set of attributes

Schema *typeGroup of attributes:*
 .../videoFormat/videoTechnicalAttributeString/@typeLabel
 or
 .../videoFormat/videoTechnicalAttributeString/@typeLink
 and (optional)
 .../videoFormat/videoTechnicalAttributeString/@typeDefinition

formatGroup of attributes:
 .../videoFormat/videoTechnicalAttributeString/@formatLabel
 or
 .../videoFormat/videoTechnicalAttributeString/@formatLink
 and (optional)
 .../videoFormat/videoTechnicalAttributeString/@formatDefinition

4.2.9.5.5 Video Technical Attribute Integer

Cardinality Multiple per Video Format

Requirement Optional

Definition Additional properties of the video resource

Format Integer with type group set of attributes

Schema typeGroup of attributes:
 .../videoFormat/videoTechnicalAttributeInteger/@typeLabel
 or
 .../videoFormat/videoTechnicalAttributeInteger/@typeLink
 and (optional)
 .../videoFormat/videoTechnicalAttributeInteger/@typeDefinition

4.2.9.5.6 Video Technical Attribute Flag

Cardinality Multiple per Video Format

Requirement Optional

Definition Additional properties of the video resource

Format Boolean with type group set of attributes

Schema typeGroup of attributes:
 .../videoFormat/videoTechnicalAttributeFlag/@typeLabel
 or
 .../videoFormat/videoTechnicalAttributeFlag/@typeLink
 and (optional)
 .../videoFormat/videoTechnicalAttributeFlag/@typeDefinition

4.2.9.5.7 Video Technical Attribute Float

Cardinality Multiple per Video Format

Requirement Optional

Definition Additional properties of the video resource



Format Float with type group set of attributes

Schema typeGroup of attributes:
`.../videoFormat/videoTechnicalAttributeFloat/@typeLabel`
or
`.../videoFormat/videoTechnicalAttributeFloat/@typeLink`
and (optional)
`.../videoFormat/videoTechnicalAttributeFloat/@typeDefinition`

4.2.9.6 Format Audio Format

Cardinality Unique per Encoding

Requirement Optional

Definition To provide information on the Audio

Format Sets of elements describing the audio

Example `<format>`
`<audioFormat>`
`<audioEncoding typeLabel="HE-AAC"/>`
`<audioTrack>`
`<audioTrackId>1</audioTrackId>`
`<audioTrackName>Narrator</audioTrackName>`
`<audioTrackLanguage typeLabel="nb"/>`
`</audioTrack>`
`</audioFormat>`
`</format>`

Refinement Audio Encoding, Audio Track Configuration, Audio Track, Audio Technical Attribute String, Audio Technical Attribute Integer, Audio Technical Attribute Flag, Audio Technical Attribute Float

4.2.9.6.1 Audio Encoding

Cardinality Unique per Audio Format

Requirement Optional

Definition The format of the audio encoding used

Format Type group set of attributes

Schema typeGroup of attributes:
`/coreMetadataType/format/audioFormat/audioEncoding/@typeLabel`
or
`/coreMetadataType/format/audioFormat/audioEncoding/@typeLink`
and (optional)
`/coreMetadataType/format/audioFormat/audioEncoding/@typeDefinition`

4.2.9.6.2 Audio Track Configuration

Cardinality Unique per Audio Format

Requirement Optional

Definition The configuration of the several audio tracks in the resource

Format String with type definition group



Schema typeGroup of attributes
 /coreMetadataType/format/audioFormat/audioTrackConfiguration/@typeLabel
 or
 /coreMetadataType/format/audioFormat/audioTrackConfiguration/@typeLink
 and (optional)
 /coreMetadataType/format/audioFormat/audioTrackConfiguration/@typeDefinition

4.2.9.6.3 Audio Track

Cardinality Multiple per Audio Format

Requirement Optional

Definition The format of the audio track(s)

Format Type group set of attributes, with additional audio track id, name and language

Schema typeGroup of attributes:
 /coreMetadataType/format/audioFormat/audioTrack/@typeLabel
 or
 /coreMetadataType/format/audioFormat/audioTrack/@typeLink
 and (optional)
 /coreMetadataType/format/audioFormat/audioTrack/@typeDefinition
 (optional)
 /coreMetadataType/format/audioFormat/audioTrack/audioTrackId
 (optional)
 /coreMetadataType/format/audioFormat/audioTrack/audioTrackName
 (optional)
 /coreMetadataType/format/audioFormat/audioTrack/audioTrackLanguage

4.2.9.6.4 Audio Technical Attribute String

Cardinality Multiple per Audio Format

Requirement Optional

Definition Additional custom properties of the audio resource referenced through an external table

Format String with type and format group set of attributes

Schema *typeGroup of attributes:*
 .../audioFormat/audioTechnicalAttributeString/@typeLabel
 or
 .../audioFormat/audioTechnicalAttributeString/@typeLink
 and (optional)
 .../audioFormat/audioTechnicalAttributeString/@typeDefinition

formatGroup of attributes:
 .../audioFormat/audioTechnicalAttributeString/@formatLabel
 or
 .../audioFormat/audioTechnicalAttributeString/@formatLink



and (optional)

.../audioFormat/audioTechnicalAttributeString/@formatDefinition

4.2.9.6.5 Audio Technical Attribute Integer

Cardinality Multiple per Audio Format

Requirement Optional

Definition Additional custom properties of the audio resource referenced through an external table

Format Integer with type group set of attributes

Schema *typeGroup of attributes:*

.../audioFormat/audioTechnicalAttributeInteger/@typeLabel

or

.../audioFormat/audioTechnicalAttributeInteger/@typeLink

and (optional)

.../audioFormat/audioTechnicalAttributeInteger/@typeDefinition

4.2.9.6.6 Audio Technical Attribute Flag

Cardinality Multiple per Audio Format

Requirement Optional

Definition Additional custom properties of the audio resource referenced through an external table

Format Boolean with type group set of attributes

Schema *typeGroup of attributes:*

.../audioFormat/audioTechnicalAttributeFlag/@typeLabel

or

.../videoFormat/videoTechnicalAttributeFlag/@typeLink

and (optional)

.../audioFormat/audioTechnicalAttributeFlag/@typeDefinition

4.2.9.6.7 Audio Technical Attribute Float

Cardinality Multiple per Audio Format

Requirement Optional

Definition Additional custom properties of the audio resource referenced through an external table

Format Float with type group set of attributes

Schema *typeGroup of attributes:*

.../audioFormat/audioTechnicalAttributeFloat/@typeLabel

or

.../audioFormat/audioTechnicalAttributeFloat/@typeLink

and (optional)

.../audioFormat/audioTechnicalAttributeFloat/@typeDefinition

4.2.9.7 Format File Format

Cardinality	Unique per Encoding
Requirement	Optional
Definition	To provide information on the File Format in complement to stream encoding information
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	<i>typeGroup of attributes:</i> <code>/coreMetadataType/format/fileFormat/@typeLabel</code> and/or <code>/coreMetadataType/format/fileFormat/@typeLink</code> and (optional) <code>/coreMetadataType/format/fileFormat/@typeDefinition</code>
Reference Data	ebu_FileFormatCS
Example	<pre> <format> <fileFormat formatLabel="mpeg" formatLink="http://www.ebu.ch/metadata/cs/ebu_FileFormatCS.xml#7.2.2" formatDefinition="The file format or wrapper defined by ISO/IEC (so called MPEG Transport Stream)"/> </format> </pre>

4.2.9.8 Format Captioning Format

Cardinality	Unique per Captioning
Requirement	Optional
Definition	To provide information on the Caption format and purpose
Format	Extended language Type
Schema	<i>typeGroup of attributes:</i> <code>/coreMetadataType/format/captioningFormat/@typeLabel</code> and/or <code>/coreMetadataType/format/captioningFormat/@typeLink</code> and (optional) <code>/coreMetadataType/format/captioningFormat/@typeDefinition</code> <i>formatGroup of attributes:</i> <code>/coreMetadataType/format/captioningFormat/@formatLabel</code> and/or <code>/coreMetadataType/format/captioningFormat/@formatLink</code> and (optional) <code>/coreMetadataType/format/captioningFormat/@formatDefinition</code> (optional) <code>/coreMetadataType/format/captioningFormat/dc:language</code> (optional) <code>/coreMetadataType/format/captioningFormat/@note</code> (optional) <code>/coreMetadataType/format/captioningFormat/captioningSourceUri</code>
Refinement	Language, Note, Captioning Source URI, typeGroup, formatGroup



4.2.9.9 Signing Captioning Format

Cardinality	Unique per Signing
Requirement	Optional
Definition	To provide information on the Signing format and purpose
Format	Extended language Type
Schema	<i>typeGroup of attributes:</i> <code>/coreMetadataType/format/signingFormat/@typeLabel</code> and/or <code>/coreMetadataType/format/signingFormat/@typeLink</code> and (optional) <code>/coreMetadataType/format/signingFormat/@typeDefinition</code> <i>formatGroup of attributes:</i> <code>/coreMetadataType/format/signingFormat/@formatLabel</code> and/or <code>/coreMetadataType/format/signingFormat/@formatLink</code> and (optional) <code>/coreMetadataType/format/signingFormat/@formatDefinition</code> (optional) <code>/coreMetadataType/format/signingFormat/dc:language</code> (optional) <code>/coreMetadataType/format/signingFormat/@note</code> (optional) <code>/coreMetadataType/format/signingFormat/captioningSourceUri</code>
Refinement	Language, Note, Captioning Source URI, typeGroup, formatGroup

4.2.9.10 Format Start

Cardinality	Unique per Medium
Requirement	Optional
Definition	The beginning point for playback of a time-based resource, such as within a digital video or audio track. Used in combination with Duration to identify a sequence or segment of a resource that has a fixed start time and end time. The start time can be expressed in different time forms inc. a timecode, normal play time, a number of edit units or user custom time references.
Format	Text string
Reference data	ANSI/SMPTE 12M-1986 (Timecode), normal play time (RFC 2326), ISO 8601, edit units (for example: samples, frames), seconds, milliseconds
Schema	<code>/coreMetadataType/format/start/</code>
Refinement	Time Format (for example RFC 3339, or Edit Unit, or Timecode, or AES TCF), either Timecode, Normal Play Time, Edit Unit Number or Time can be used

4.2.9.10.1 Format Start Timecode

Cardinality	Unique per Medium
Requirement	Optional
Definition	The timecode of the start



Format	Text string
Reference data	ANSI/SMPTE 12M-1986 (Timecode)
Schema	<code>/coreMetadataType/format/start/timecode</code>
Example	<pre> <format> <start> <timecode>23:45:14:14</timecode> </start> </format> </pre>

4.2.9.10.2 Format Start Normal Play Time

Cardinality	Unique per Medium
Requirement	Optional
Definition	The time of the start
Format	Text string
Reference data	Normal play time (RFC 2326), ISO 8601
Schema	<code>/coreMetadataType/format/start/normalPlayTime</code>
Example	<pre> <format> <start> <time>23:45:14</time> </start> </format> </pre>

4.2.9.10.3 Format Start Edit Unit Number

Cardinality	Unique per Medium
Requirement	Optional
Definition	The Edit Unit count of the first sample at the start. [check that "sample" is used correctly throughout the document] The Edit Unit is either the fraction of a second calculated as an inverse of the editRate (video frame rate or audio sample rate) of the resource, possibly corrected by a factor provided in the denominator and numerator attributes, or the smallest amount of time per unit (for example, a second or millisecond). $\text{editUnit} = 1 / (\text{editRate} * (\text{factorNumerator} / \text{factorDenominator}))$ The start time is in this case an integer indicating a number of Edit Units, that is the corresponding editUnitNumber.
Format	edit units (for example: samples, frames), seconds, milliseconds
Schema	<pre> /coreMetadataType/format/start/ editUnitNumber / /coreMetadataType/format/start/ editUnitNumber/@editRate /coreMetadataType/format/start/ editUnitNumber/@factorNumerator /coreMetadataType/format/start/ editUnitNumber/@factorDenominator </pre>
Example	<pre> <format> <start> <editUnitNumber editRate="48000">384000</editUnitNumber> </start> </format> OR <format> <start> </pre>



```

    <editUnitNumber editRate="30" factorNumerator="1000"
    factorDenominator="1001">120</editUnitNumber>
  </start>
</format>

```

4.2.9.10.4 Format Start Time

Cardinality	Unique per Medium
Requirement	Optional
Definition	String with typeGroup attributes to describe the time, used for old archives
Format	String
Schema	<pre> /coreMetadataType/format/start/time /coreMetadataType/format/start/time/@typeLabel /coreMetadataType/format/start/time/@typeLink /coreMetadataType/format/start/time/@typeDefinition </pre>
Example	<pre> <format> <start> <time>A quarter past Nine</time> </start> </format> </pre>

4.2.9.11 Format Duration

Cardinality	Unique per physical realisation of an item
Requirement	Optional
Definition	Time-based duration (extent) of the resource. The duration can be expressed in different time forms including a timecode or a number of edit units. The Edit Unit is either the fraction of a second calculated as an inverse to the frame (video) or sample (audio) rate of the resource, or the smallest amount of time per unit (for example: a second or millisecond). The duration is in this case an integer indicating a number of Edit Units.
Format	Text string
Reference data	ANSI/SMPTE 12M-1986 (Timecode), edit units (for example: samples, frames), seconds, milliseconds), sample count expressed as AES TCF, normal play time (RFC 2326), ISO 8601
Schema	<pre>/coreMetadataType/format/duration</pre>
Refinement	Time Format (for example: RFC 3339 or Edit Unit or Timecode)

4.2.9.11.1 Format Duration Timecode

Cardinality	Unique per Medium
Requirement	Optional
Definition	The timecode of the duration
Format	Text string
Ref. data	ANSI/SMPTE 12M-1986 (Timecode)
Schema	<pre>/coreMetadataType/format/duration/timecode</pre>
Example	<pre><format></pre>



```

<duration>
  <timecode>00:45:10:04</timecode>
</duration>
</format>

```

4.2.9.11.2 Format Duration Normal Play Time

Cardinality	Unique per Medium
Requirement	Optional
Definition	The time of the duration
Format	Text string
Ref. data	Normal play time (RFC 2326), ISO 8601
Schema	/coreMetadataType/format/duration/normalPlayTime
Example	<pre> <format> <duration> <normalPlayTime>PT45M10S</normalPlayTime> </duration> </format> </pre>

4.2.9.11.3 Format Duration Edit Unit Number

Cardinality	Unique per Medium
Requirement	Optional
Definition	The sample-accurate sample of the duration. The Edit Unit is either the fraction of a second calculated as an inverse of the editRate (video frame rate or audio sample rate) of the resource, possibly corrected by a factor provided in the denominator and numerator attributes, or the smallest amount of time per unit (for example, a second or millisecond). $\text{editUnit} = 1 / (\text{editRate} * (\text{factorNumerator} / \text{factorDenominator}))$ The start time is in this case an integer indicating a number of Edit Units; that is, the corresponding editUnitNumber.
Format	edit units (for example: samples, frames), seconds, milliseconds
Schema	<pre> /coreMetadataType/format/duration/ editUnitNumber/ /coreMetadataType/format/duration/ editUnitNumber/@editRate /coreMetadataType/format/duration/ editUnitNumber/@factorNumerator /coreMetadataType/format/duration/ editUnitNumber/@factorDenominator </pre>
Example	<pre> <format> <duration> <editUnitNumber editRate="1000">180000</editUnitNumber> </duration> </format> </pre>

4.2.9.11.4 Format Duration Time

Cardinality	Unique per Medium
Requirement	Optional
Definition	String with typeGroup attributes to describe the duration
Format	String
Schema	<pre> /coreMetadataType/format/duration/time /coreMetadataType/format/duration/time/@typeLabel </pre>



Example	<pre> /coreMetadataType/format/duration/time/@typeLink /coreMetadataType/format/duration/time/@typeDefinition <format> <duration> <time>1'34"</time> </duration> </format> </pre>
4.2.9.12	Format File Size
Cardinality	Unique per Encoding
Requirement	Optional
Definition	To indicate the storage requirements or file size of a digital resource. The file size is expressed in bytes.
Format	Non-negative integer
Schema	/coreMetadataType/format/fileName
Example	<pre> <format> <fileSize>234572</fileSize> </format> </pre>
4.2.9.13	Format Filename
Cardinality	Unique per Medium
Requirement	Optional
Definition	The name of the file as it appears in the location path or url.
Format	Free text
Schema	/coreMetadataType/format/filename
Example	<pre> <format> <filename>Content.mp4</filename> </format> </pre>
4.2.9.14	Format Locator
Cardinality	Unique per medium
Requirement	Optional
Definition	An "address for a resource". For an organisation or producer acting as caretaker for a media resource, Format Location may contain information about a specific (for example) tape name, shelf location for an asset, including an organisation's name, departmental name, shelf id. and contact information. The Format Location for a data file or web page may include a complete URI with a domain, path, filename or html URL.
Format	Free text
Schema	/coreMetadataType/format/locator
Reference data	URI: Unique Resource Identifier, URL: Unique Resource Locator: http://tools.ietf.org/html/rfc3986
Example	<pre> <format> <locator>Archives Building A, Row J, Shelf 2</locator> </format> OR <format> <locator>http://www.ebu.ch/CorporateVideo.avi</locator> </format> </pre>

4.2.9.15 Technical Attribute String

Cardinality	Multiple per Format
Requirement	Optional
Definition	Additional custom properties of the resource referenced through an external table
Format	String with type and format group set of attributes
Schema	typeGroup of attributes: /coreMetadataType/format/technicalAttributeString/@typeLabel or /coreMetadataType/format/technicalAttributeString/@typeLink and (optional) /coreMetadataType/format/technicalAttributeString/@typeDefinition formatGroup of attributes: /coreMetadataType/format/technicalAttributeString/@formatLabel or /coreMetadataType/format/technicalAttributeString/@formatLink and (optional) /coreMetadataType/format/technicalAttributeString/@formatDefinition

4.2.9.16 Technical Attribute Integer

Cardinality	Multiple per Format
Requirement	Optional
Definition	Additional custom properties of the resource referenced through an external table
Format	Integer with type group set of attributes
Schema	typeGroup of attributes: /coreMetadataType/format/technicalAttributeInteger/@typeLabel or /coreMetadataType/format/technicalAttributeInteger/@typeLink and (optional) /coreMetadataType/format/technicalAttributeInteger/@typeDefinition

4.2.9.17 Technical Attribute Flag

Cardinality	Multiple per Format
Requirement	Optional
Definition	Additional custom properties of the resource referenced through an external table
Format	Boolean with type group set of attributes
Schema	typeGroup of attributes: /coreMetadataType/format/technicalAttributeFlag/@typeLabel or /coreMetadataType/format/technicalAttributeFlag/@typeLink



and (optional)

/coreMetadataType/format/technicalAttributeFlag/@typeDefinition

4.2.9.18 Technical Attribute Float

Cardinality Multiple per Format

Requirement Optional

Definition Additional custom properties of the resource referenced through an external table

Format Float with type group set of attributes

Schema typeGroup of attributes:

/coreMetadataType/format/technicalAttributeFloat/@typeLabel

or

/coreMetadataType/format/technicalAttributeFloat/@typeLink

and (optional)

/coreMetadataType/format/technicalAttributeFloat/@typeDefinition

4.2.10 Identifier

Cardinality Multiple

Requirement Mandatory

Definition A unique, unambiguous reference or identifier for a resource within a given context. Best practice is to identify the resource (whether analogue or digital) by means of a string or number corresponding to an established or formal identification system if one exists. Otherwise, use an identification method that is in use within your agency, station, production company, office, or institution.

It is also possible to enter more than one, different but still unique, identifier for the same resource.

Format Free text

Schema **/coreMetadataType/identifier/dc:identifier**

Example **<identifier>**

<dc:identifier>06.0A.2B.34.01.01.01.01</dc:identifier>

</identifier>

Refinement Identifier Type, Identifier Format, Identifier Note (optional additional contextual information)

4.2.10.1 Identifier Type

Cardinality Unique per Identifier

Requirement Optional

Definition Used to define the type of Identifier used.

Format Classification scheme link (reference data file URL plus an optional classification scheme term code)

or

free text (label) and an optional definition

Schema *typeGroup of attributes:*

/coreMetadataType/identifier/@typeLabel



and/or
 /coreMetadataType/identifier/@typeLink
 and (optional)
 /coreMetadataType/identifier/@typeDefinition

Example <identifier typeLabel="Main" typeDefinition="main identifier attributed to the resource">
 <dc:identifier>06.0A.2B.34.01.01.01.01</dc:identifier>
 </identifier>

4.2.10.2 Identifier Format

Cardinality Unique per Identifier

Requirement Optional

Definition Used in combination with the resource Identifier. It can denote the agency or institution who specified or assigned it.

Format Classification scheme link (reference data file URL plus an optional classification scheme term code)

or

free text (label).

Schema *formatGroup of attributes:*

/coreMetadataType/identifier/@formatLabel

or

/coreMetadataType/identifier/@formatLink

and (optional)

/coreMetadataType/identifier/@formatDefinition

Reference data ebu_IdentifierTypeCodeCS

URI - RFC3986

Example <identifier formatLabel="SMPTE Unique Material Identifier (UMID)"
 formatLink="http://www.ebu.ch/metadata/cs/ebu_IdentifierTypeCodeCS.xml#
 1.1" formatDefinition="a unique identifier as defined by SMPTE 330M"
 typeLabel="Main" typeDefinition="main identifier attributed to the
 resource">
 <dc:identifier>06.0A.2B.34.01.01.01.01</dc:identifier>
 </identifier>

4.2.11 Source

Cardinality Multiple

Requirement Optional

Definition Reference to the resource(s) from which the current resource is derived in whole or in part.

If no label or number is available, the title and/or the statement of responsibility etc. of the digitized recording is recorded here. For a digitized radio programme the production number is normally given here.

The Recommended best practice is to use a unique identifier to identify the physical source that has been used to create the digital resource. In the case of a digitized analogue recording, it is the recording used for digitization which is the source. For commercial recordings the label and number is normally given here.

Format Free text

Schema /coreMetadataType/dc:source



Example `<dc:source> Eurovision feed 2007-07-16T19:20:30.45+01:00</dc:source>`

4.2.12 Language

Cardinality Multiple

Requirement Optional

Definition Identifies languages and their use in the intellectual content of the resource. Recommended best practice for the values of the Language element is defined by RFC 1766, which includes a two-letter Language Code (taken from the ISO Standard 639), followed optionally, by a two-letter Country Code (taken from the ISO Standard 3166). For example, 'en' for English, 'fr' for French, or 'en-uk' for English used in the United Kingdom.

More contextual information can be provided using the “note” attribute.

Format Free text or code

Schema `/coreMetadataType/language/dc: language`
or

`/coreMetadataType/language/languageCode`

Reference data `ebu_Iso639_1LanguageCodeCS`, `ebu_Iso639_2LanguageCodeCS`,
`ebu_Iso3166CountryCodeCS`

ISO 639-1 or ISO 639-2 optionally combined with an ISO 3166 country code, in compliance with RFC 3066

Example `<language>`
`<dc: language>FR</dc: language>`
`</language>`

Refinement Language type, note

4.2.12.1 Language Type

Cardinality Unique per Language

Requirement Optional

Definition Indicates the usage/purpose of the language described by the Language element.

Format Classification scheme link (reference data file URL plus an optional classification scheme term code)

or

free text (label) and an optional definition

Schema **typeGroup of attributes:**

`/coreMetadataType/language/@typeLabel`

or

`/coreMetadataType/language/@typeLink`

and (optional)

`/coreMetadataType/language/@typeDefinition`

Reference data `ebu_LanguagePurposeCodeCS`

Example `<language typeLabel="Main original language"`
`typeLink="http://www.ebu.ch/metadata/cs/ebu_LanguagePurposeCodeCS.xml#1`
`.1">`
`<dc: language>FR</dc: language>`
`</language>`

4.2.13 Relation

Cardinality	Multiple per relation
Requirement	Optional
Definition	Recommended best practice is to reference the resource (to which the current resource under description is related) by means of a string or number conforming to a formal identification system. Relation is used to show the relation in content to another resource. For example, "IsPartOf" is used to show the relation between a part of a radio programme and the whole programme or between a track and a record album. A resource can be identified by its title, or an identifier (possibly a URI). The related item has its own separate metadata record. Relation is used to provide a name, an identification number or ID, or a locator where the related item can be found.
Format	String, identifier or URI
Schema	<pre> /coreMetadataType/relation/dc:relation /coreMetadataType/relation/relationIdentifier /coreMetadataType/relation/relationLink </pre>
Example	<pre> <isPartOf> <relationIdentifier> <dc:identifier>06.0A.2B.30.01.01.01.01</dc:identifier> </relationIdentifier> </isPartOf> </pre>
Refinement	Relation Type, Relation Format, Running Order Flag

4.2.13.1 Relation Type

Cardinality	Unique per Relation
Requirement	Optional
Definition	To show the type of relation to another resource, for example, identifies ways in which the resource is related by intellectual content to some other resource. The relation type shall be used if none of the following predefined relations can be used: isVersionOf or hasVersion isReplacedBy or replaces isRequiredBy or requires isPartOf or hasPart isReferencedBy or references isFormatOf or hasFormat
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	<i>typeGroup of attributes:</i> <pre> /coreMetadataType/relation/@typeLabel and/or /coreMetadataType/relation/@typeLink and (optional) </pre>



	<code>/coreMetadataType/relation/@typeDefinition</code>
Reference data	<code>tva_HowRelatedCS</code>
Example	<pre> <relation typeLabel="isRemakeOf"> <dc:relation>Mathas küche</dc:relation> </relation> <isVersionOf> <relationIdentifier> <dc:identifier>06.0A.2B.34.01.01.01.01</dc:identifier> </relationIdentifier> </isVersionOf> </pre>
4.2.13.2 Running Order Number	
Cardinality	Unique
Requirement	Optional
Definition	If used, optional field to indicate that the relation is hierarchical and that there is an order in which content is chronologically related, indicates the running order of an element
Format	integer
Schema	<code>/assetDescriptivecoreMetadataMetadataType/relation/@runningOrderNumber</code>
Example	<pre> <isVersionOf> <relationIdentifier> <dc:identifier>06.0A.2B.34.01.01.01.01</dc:identifier> </relationIdentifier> </isVersionOf> </pre>

4.2.14 Coverage

Cardinality	Unique
Requirement	Optional
Definition	Coverage is used to show various time and place aspects of the subject of the content. Coverage will typically include spatial location (a place name or geographic coordinates), temporal period (a period label, date, or date range) or jurisdiction (such as a named administrative entity). Recommended best practice is to select a value from a controlled vocabulary (for example, the Thesaurus of Geographic Names) and that, where appropriate, named places or time periods be used in preference to numeric identifiers such as sets of coordinates or date ranges.
Format	Free text
Schema	<code>/coreMetadataType/coverage/dc:coverage</code>
Example	<pre> <coverage> <dc:coverage>London, 15th Century</dc:coverage> </coverage> </pre>
Refinement	Coverage Temporal, Coverage Spatial

4.2.14.1 Coverage Temporal

Cardinality	Unique per Coverage
Requirement	Optional
Definition	Temporal characteristics of the content of the resource. To indicate (for example) dates, times or periods specific to the resource in complement to Description.
Format	Free text



Schema	<code>/coreMetadataType/coverage/temporal</code>
Reference data	ISO 8106, W3C-DTF (Date and Time Format)
Refinement	Temporal Period of Time, Temporal Type

4.2.14.1.1 Temporal Period of Time

Cardinality	Unique per Period of Time
Requirement	Optional
Definition	To provide detailed information on the time boundaries and period of time related to the resource.
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition.
Schema	<i>dateGroup of attributes</i> <code>/coreMetadataType/coverage/temporal/periodOfTime/@startDate</code> and/or <code>/coreMetadataType/coverage/temporal/periodOfTime/@endDate</code> and (optional) <code>/coreMetadataType/coverage/temporal/periodOfTime/@period</code>
Example	<pre><coverage> <dc:coverage>London, 15th Century</dc:coverage> <temporal> <PeriodOfTime startYear="1410" endYear="1490"/> </temporal> </coverage></pre>

4.2.14.1.2 Temporal Type

Cardinality	Unique per Temporal
Requirement	Optional
Definition	To precise the type of temporal information provided.
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	<i>typeGroup of attributes:</i> <code>/coreMetadataType/coverage/temporal/@typeLabel</code> and/or <code>/coreMetadataType/coverage/temporal/@typeCode</code> and (optional) <code>/coreMetadataType/coverage/temporal/@typeDefinition</code>



Example `<coverage>
 <dc:coverage>London, 15th Century</dc:coverage>
 <temporal typeLabel="Fictional action date" typeDefinition="the date
 at which the event presented is supposed to take place">
 <PeriodOfTime startYear="1410" endYear="1490"/>
 </temporal>
</coverage>`

4.2.14.2 Coverage Spatial

Cardinality Unique per Coverage

Requirement Optional

Definition Spatial characteristics of the content of the resource.

Schema `/coreMetadataType/coverage/spatial`

Refinement Location Name

4.2.14.2.1 Location Name

Cardinality Multiple

Requirement Optional

Definition To indicate, for example, specific place and location aspects of the resource in complement to Description.

Format Classification scheme (code),
free text

or the latitude and longitude expressed in decimal degrees

Schema `/coreMetadataType/coverage/spatial/location/Name`

or

`/coreMetadataType/coverage/spatial/locationName/Code`

or

`{/coreMetadataType/coverage/spatial/locationName/posx
and
/coreMetadataType/coverage/spatial/locationName/posy}`

Ref. data `ebu:UNTerritoryCodeCS, ebu:Iso3166CountryCodeCS.`

Thesaurus of Geographic Names, latitude and longitude coordinates

Example `<spatial>
 <location>
 <name>London</name>
 <posy>51.49</posy>
 <posx>-0.15</posx>
 <code>W1AA 4WW</code>
 </location>
</spatial>`

Refinement Location Name Type, Location Name Note (optional additional contextual information)

4.2.14.2.2 Location Name Type

Cardinality Unique per Location Name

Requirement Optional

Definition To precise the type of place and location.



Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or free text (label) and an optional definition
Schema	typeGroup of attributes: /coreMetadataType/coverage/spatial/locationName/@typeLabel and/or /coreMetadataType/coverage/spatial/locationName/@typeCode and (optional) /coreMetadataType/coverage/spatial/locationName/@typeDefinition
Example	<pre><spatial> <location typeLabel="City" typeDefinition="to provide the name of a city"> <name>London</name> <posy>51.49</posy> <posx>-0.15</posx> <code>W1AA 4WW</code> </location> </spatial></pre>

4.2.15 Rights

Cardinality	Multiple
Requirement	Optional
Definition	An all-purpose field to identify information (rights management statement or reference to a service providing such information, for example via a URL) about copyright, intellectual property rights or other property rights held in and over a resource, stating whether access is open or restricted in some way. If dates, times, territories and availability periods are associated with a right, they should be included. If the Rights element is absent, no assumptions can be made about the status of these and other rights with respect to the resource.
Format	Free text, URL
Schema	/coreMetadataType/rights/dc:rights or /coreMetadataType/rights/rightsLink
Example	<pre><rights> <dc:rights>Copyright limited to United Kingdom satellite broadcast delivery</dc:rights> </rights></pre>
Refinement	Rights Type, Rights Holder, Coverage, Exploitation Issues

4.2.15.1 Rights Type

Cardinality	Unique per Rights
Requirement	Optional
Definition	Used to define the type of rights expressed
Format	Classification scheme link (reference data file URL plus an optional classification scheme term code) or



	free text (label) and an definition (optional)
Schema	typegroup of attributes: <pre> /coreMetadataType/rights/@typeLabel and/or /coreMetadataType/rights/@typeCode and (optional) /coreMetadataType/rights/@typeDefinition </pre>
Example	<pre> <rights typeLabel="Publishing rights"> <dc:rights>Rights for web-publishing</dc:rights> <rightsHolder> <contactDetails> <name> <name>BBC</name> </name> </contactDetails> </rightsHolder> <exploitationIssues>Can be published one week after broadcast, geoblocked to Norway</exploitationIssues> <coverage> <temporal> <PeriodOfTime startDate="2010-10-10" endDate="2010-10-17"/> </temporal> <spatial> <location> <name>Norway</name> </location> </spatial> </coverage> <rightsId> <dc:identifier>Contract-45-2010-b</dc:identifier> </rightsId> </rights> </pre>

4.2.15.2 Rights Holder

Cardinality	Multiple
Requirement	Optional
Definition	A person or organisation owning or managing the rights of the resource.
Format	AES Core Entity Type (see 4.2.19)
Schema	<pre> /coreMetadataType/creator/entity /coreMetadataType/rights/rightsHolder </pre>
Example	See 4.2.15.1
Refinement	Entity (Contact Details, Organisation Details, Role)

4.2.15.3 Coverage

Cardinality	Unique per Rights description
Requirement	Optional
Definition	Specifies a specific start date, end date, or period for the availability of the item or the date from which the rights or exploitation issues apply. It may refer to start dates for the availability of an item that is used within a particular geographical area; for example, broadcast locally, regionally, nationally or internationally, or for web-based distribution. A specific time may



also be associated with the date.

Format	Text string
Schema	/coreMetadataType/rights/coverage
Reference data	ISO 8601, W3C-DTF (Date & Time Format)
Example	See 4.2.15.1

4.2.15.4 Exploitation Issues

Cardinality	Multiple
Requirement	Optional
Definition	Use to state any other restrictions, such as non-rights ones; for example, legal. State by media, territory, scope (restriction on whole item or extracts) and possibly language. The presence of this information can be used by asset management system implementing traffic lights like mechanism to signal that content may be subject to particular restrictions to be clarified before exploitation.
Format	Free text
Schema	/coreMetadataType/rights/exploitationIssues
Example	Legal restriction, editorial embargo, etc., (see 4.2.15.1)

4.2.16 Version

Cardinality	Unique
Requirement	Optional
Definition	Expresses the version type.
Format	Enumerated list, classification scheme or free text
Schema	/coreMetadataType/version
Example	<version>US Version</version>
Mapping	'Version' is not a Dublin Core element but would be mapped to dc:description

4.2.17 Publication History

Cardinality	Unique
Requirement	Optional
Definition	To provide information about the publication history.
Schema	/coreMetadataType/publicationHistory
Refinement	First Transmission Date/Time, First Transmission Channel, Repeat Date/Time, Repeat Channel

4.2.17.1 First Transmission Date and Time

Cardinality	Unique
Requirement	Optional
Definition	The first transmission date
Format	Text string
Schema	/coreMetadataType/publicationHistory/firstTransmissionDateTime



Reference data RFC 3339 (dateTime).
Note that RFC 3339 is more strict than the W3C Date and Time Format. In particular, all portions of the date and time must be present, with the exception of the fractional portions of the second, which may be omitted.

Example

```
<publicationHistory>
  <firstTransmissionDate>2010-04-15</firstTransmissionDate>
  <firstTransmissionTime>16:00:00</firstTransmissionTime>
  <firstTransmissionChannel>SWR4</firstTransmissionChannel>
  <repeatDate>2010-05-02</repeatDate>
  <repeatTime>22:00:00</repeatTime>
  <repeatChannel>SWR4</repeatChannel>
  <repeatDate>2010-05-03</repeatDate>
  <repeatTime>22:30:00</repeatTime>
  <repeatChannel>NDR2</repeatChannel>
</publicationHistory>
```

4.2.17.2 First Transmission Channel

Cardinality Unique
Requirement Optional
Definition The channel on which the title was first transmitted
Format Free text
Schema /coreMetadataType/publicationHistory/firstTransmissionChannel
Example See 4.2.17.1

4.2.17.3 Repeat Date and Time

Cardinality Multiple
Requirement Optional
Definition Subsequent transmission date and time
Format Text string
Schema /coreMetadataType/publicationHistory/repeatDateTime
Reference data RFC 3339 (dateTime).
Note that RFC 3339 is more strict than the W3C Date and Time Format. In particular, all portions of the date and time must be present, with the exception of the fractional portions of the second, which may be omitted.
Example See 4.2.17.1

4.2.17.4 Repeat Channel

Cardinality Unique per Repeat Date / Time
Requirement Optional
Definition The channel on which the title was subsequently transmitted
Format Free text
Schema /coreMetadataType/publicationHistory/repeatChannel
Example See 4.2.17.1



4.2.18 Metadata Provider

Cardinality	Unique per asset management description
Requirement	Required
Definition	Identifies the metadata provider, for example the contributing archive. The organisation ID or name provides the archive ID or name required for OAI metadata harvesting operation.
Format	AES Core Entity Type (see 4.2.19) /coreMetadataType/creator/entity
Schema	/coreMetadataType/metadataProvider

4.2.19 Entity Type

Cardinality	Unique
Requirement	Optional
Definition	Provides details information about a person, a group of persons, or organisation
Schema	/coreMetadataType/creator/entity /coreMetadataType/contributor/entity /coreMetadataType/publisher/entity /coreMetadataType/rights/rightsOwner/entity /coreMetadataType/rights/metadataProvider/entity
Refinement	Contact Details, Organisation Details, Role

4.2.19.1 Contact Details

Cardinality	Unique per Entity
Requirement	Optional
Definition	Minimum information providing means to further identify and contact the entity.
Format	Contact Details Type (see schema in Annex F): name: e-mail, occupation, stage name, contacts
Refinement	Contact Id, Name, Contact Email Address, Contact Occupation, Stage Name, Related Contacts (entity)

4.2.19.2 Organisation Details

Cardinality	Unique per Entity
Requirement	Optional
Definition	Minimum information providing means to further identify and contact the entity.
Format	Organisation Details Type (see schema in Annex F): name, department, web address, contacts
Refinement	Organisation ID, Organisation Name, Organisation Department, Organisation Web Address, Contacts (entity) [Expand descriptions]

4.2.19.3 Role

Cardinality	Unique per Subject Type
Requirement	Optional
Definition	Used to identify the function fulfilled by the person, group or organisation described as an entity.



Format	Enumerated list, classification scheme or free text
Ref. data	ebu_RoleCodeCS
Example	Director



Annex A: (Normative) AES Core Metadata Set Schema

The AES Core metadata schema is used to generate AES Core Metadata instances formed of an **aesCoreMain** document.

The **aesCoreMain** document contains several attributes required to contribute to the OAI (Open Archive Initiative) for metadata harvesting. These attributes include the name of the schema (in case the schema location urn would not be present), the version of the schema used to generate the document, the date of last modification of the document and a unique identifier associated with the document. The name of the contributing archive is given by the metadata provider's organisation name or ID.

Resource-related information is provided by the **coreMetadata** element.

AES Core XML schema

The AES Core XML schema associated with this document is available electronically at:

<http://www.aes.org/standards/schemas/aes60-2011.xsd>

See also Annex F

Annex B: (Normative) Reference files

Document description	Contents
Schemas	http://www.aes.org/schemas/aes-core-20100617.xsd , http://www.aes.org/schemas/xml.xsd http://www.aes.org/schemas/simpliedc20021212-UK.xsd [*]

*The Simple Dublin Core Schema modified to set the language by default to UK-English instead of US-English.

Annex C: (Informative) Reference data links

Reference data alias	Link
ebu_TitleStatusCodeCS	www.ebu.ch/metadata/cs/web/ebu_TitleStatusCodeCS_p.xml.htm
ebu_RoleCodeCS	www.ebu.ch/metadata/cs/web/ebu_RoleCodeCS_p.xml.htm
ebu_IPTCSubjectCodeCS	www.ebu.ch/metadata/cs/web/ebu_IPTCSubjectCodeCS_p.xml.htm
ebu_DescriptionTypeCodeCS	www.ebu.ch/metadata/cs/web/ebu_DescriptionTypeCodeCS_p.xml.htm
ebu_ContentAlertSchemeCodeCS	www.ebu.ch/metadata/cs/web/ebu_ContentAlertSchemeCodeCS_p.xml.htm
ebu_ContentGenreCS	www.ebu.ch/metadata/cs/web/ebu_ContentGenreCS_p.xml.htm
ebu_EditorialFormatCodeCS	www.ebu.ch/metadata/cs/web/ebu_EditorialFormatCodeCS_p.xml.htm
ebu_IntentionCodeCS	www.ebu.ch/metadata/cs/web/ebu_IntentionCodeCS_p.xml.htm
tva_ContentCommercialCS	www.ebu.ch/metadata/cs/web/tva_ContentCommercialCS_p.xml.htm
tva_ContentAlertCS	www.ebu.ch/metadata/cs/web/tva_ContentAlertCS_p.xml.htm
ebu_IntendedAudienceCodeCS	www.ebu.ch/metadata/cs/web/ebu_IntendedAudienceCodeCS_p.xml.htm
ebu_ObjectTypeCS	www.ebu.ch/metadata/cs/web/ebu_ObjectTypeCS_p.xml
ebu_StorageMediaTypeCode	www.ebu.ch/metadata/cs/web/ebu_StorageMediaTypeCode_p.xml.htm
ebu_MediaTypeCS	www.ebu.ch/metadata/cs/web/ebu_MediaTypeCS_p.xml.htm
ebu_FileFormatCS	www.ebu.ch/metadata/cs/web/ebu_FileFormatCS_p.xml.htm
ebu_IdentifierTypeCodeCS	www.ebu.ch/metadata/cs/web/ebu_IdentifierTypeCodeCS_p.xml.htm
ebu_LanguagePurposeCodeCS	www.ebu.ch/metadata/cs/web/ebu_LanguagePurposeCodeCS_p.xml.htm
tva_HowRelatedCS	www.ebu.ch/metadata/cs/web/tva_HowRelatedCS_p.xml.htm

Annex D: (Normative) Maintenance

The AES Core Metadata Set is maintained by the AES Standards Committee. Suggestions for corrections or additions can be made by mailing to **standards@aes.org** where they will be considered by the relevant working group for inclusion in the next revision of this standard.



Annex E: (Informative) Bibliography

AES-R9-2008: AES standards project report - Considerations for standardising AES metadata sets, Audio Engineering Society, New York, NY, US. www.aes.org/publications/standards/

EBU Technical Information I36-2003 - Metadata Implementation considerations for Broadcasters. <http://tech.ebu.ch/publications/i036>

EBU Tech 3293-v1.2-2010 - EBU Core Metadata Set (EBUCore)

EBU Tech 3295 - P-META Metadata Library

PBCore.org (Public Broadcasting Metadata Dictionary Project Project), www.pbcore.org/index.html

OAI, The Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH), www.openarchives.org/, www.openarchives.org/OAI/openarchivesprotocol.html

ISO 639, Codes for the representation of names of languages: <http://www.loc.gov/standards/iso639-2/>. International Standards Organisation, Geneva, Switzerland. www.iso.org/

ISO 3166-1, Codes for the representation of names of countries and their subdivisions, International Standards Organisation, Geneva, Switzerland. www.iso.org/

ISO 4217, Codes for the representation of currencies and funds, International Standards Organisation, Geneva, Switzerland. www.iso.org/

ISO 8601, Data elements and interchange formats - Information interchange - Representation of dates and times, International Standards Organisation, Geneva, Switzerland. www.iso.org/

SMPTE ST12 Time and Control Code, Society of Motion Picture and Television Engineers, NY., US. www.smpte.org/standards/

SMPTE ST330 Unique Material Identifier (UMID), Society of Motion Picture and Television Engineers, NY., US. www.smpte.org/standards/

IETF RFC 2326 Real Time Streaming Protocol (RTSP), Internet Society, www.ietf.org/

IETF RFC 3066 Tags for the Identification of Languages, Internet Society, www.ietf.org/

W3C Date and Time Formats (DTF), <http://www.w3.org/TR/NOTE-datetime>. www.w3.org/

MIME Type, Internet Assigned Numbers Authority, www.iana.org/assignments/media-types/

Library of Congress Subject Heading (LCSH), <http://authorities.loc.gov/>

Library of Congress Classification (LCC), www.loc.gov/aba/cataloging/classification/

Medical Subject Headings (MeSH), www.nlm.nih.gov/mesh/

Dewey Decimal Classification (DDC), www.oclc.org/dewey/



Dansk decimalklassedeling 5.utgave (DK5), www.statsbiblioteket.dk/

Klassifikasjonssystem för svenska bibliotek (SAB), www.biblioteksforeningen.org/

Universal Decimal Classification (UDC), www.udcc.org/

Norske emneord, Oslo: Biblioteksentralen, 1996

NOTE Internet links, where given, are for convenience only. They were checked to be current at the time of publication, but may have changed since then.

Annex F: (Informative) Schema listing

NOTE see annex A for normative reference

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns="urn:ebu:metadata-
schema:ebuCore_2010" xmlns:dc="http://purl.org/dc/elements/1.1/" targetNamespace="urn:ebu:metadata-
schema:ebuCore_2010" elementFormDefault="qualified" attributeFormDefault="unqualified">
  <xs:import namespace="http://www.w3.org/XML/1998/namespace" schemaLocation="xml.xsd"/>
  <xs:import namespace="http://purl.org/dc/elements/1.1/"
    schemaLocation="simpledc20021212.xsd"/>
  <xs:element name="ebuCoreMain" type="ebuCoreMainType"/>
  <xs:annotation>
    <xs:documentation> The following Creative Commons Rights apply to the use of this EBU resource:
      http://creativecommons.org/licenses/by-nc-sa/3.0/ Owner: EBU Technical Department, Author: Jean-Pierre
      Evain Contact: evain@ebu.ch </xs:documentation>
  </xs:annotation>
  <xs:complexType name="ebuCoreMainType">
    <xs:annotation>
      <xs:documentation> ebuCoreMain is the body of a document using the
        EBU Core metadata set
      </xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="coreMetadata" type="coreMetadataType"/>
      <xs:element name="metadataProvider" type="entityType" minOccurs="0">
        <xs:annotation>
          <xs:documentation>Identifies the metadata provider, e.g. the contributing archive.
            The organisation Id or name provide the archive ID or name required for OAI
            metadata harvesting operation. </xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="schema" default="ebuCoreMetadataSet_20100820.xsd">
      <xs:annotation>
        <xs:documentation> The name of the schema for OAI management. </xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="version" default="1.2">
      <xs:annotation>
        <xs:documentation> The version of the schema for OAI management.
        </xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="dateLastModified" type="xs:date">
      <xs:annotation>
        <xs:documentation> The date of edition of the metadata instance for OAI management
        </xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="documentId" type="xs:NMTOKEN">
      <xs:annotation>
        <xs:documentation> The unique Identifier of the metadata instance for OAI management
        </xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
```

```

<xs:complexType name="coreMetadataType">
  <xs:annotation>
    <xs:documentation>The document containing all the core information regarding the
      resource</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:sequence>
      <xs:element name="title" type="titleType" minOccurs="0" maxOccurs="unbounded">
        <xs:annotation>
          <xs:documentation>A Title is the 'main' name given to a resource e.g. a media
            item, a media object, or a sequence as specified by the associated title
            type. It corresponds for a series to the series title, for a programme
            to the programme title, for an item to the item title, etc. Titles are
            recorded as they appear. The Title is the name by which a resource is
            formally known and that everyone should use to refer to or search for
            that particular resource. The Title may be provided in several
            languages. If present, the attributionDate attribute indicates when the
            Title was attributed. A Title is the 'main' name given to a resource
            e.g. a media item, a media object, or a sequence as specified by the
            associated title type. It corresponds for a series to the series title,
            for a programme to the programme title, for an item to the item title,
            etc. Titles are recorded as they appear. The Title is the name by which
            a resource is formally known and that everyone should use to refer to or
            search for that particular resource. The Title may be provided in
            several languages. If present, the attributionDate attribute indicates
            when the Title was attributed. </xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="alternativeTitle" type="alternativeTitleType"
          minOccurs="0" maxOccurs="unbounded">
          <xs:annotation>
            <xs:documentation>An Alternative Title is the name other than the 'main' Title
              given to a resource. The type of title is defined by the typeGroup of
              attributes. The status of the title is defined by the statusGroup of
              attributes. Alternative Titles are recorded as they appear. An
              Alternative Title may be attributed to a resource for several reasons
              described using the status (e.g. working title) and type (e.g. series
              title) attributes. The alternativeTitle may be provided in several
              languages. It is sometimes common practice to put dates into the
              alternativeTitle. If present, the attributionDate (indicating when the
              alternativeTitle was attributed) in the date attribute should be
              consistent. </xs:documentation>
            </xs:annotation>
          </xs:element>
        </xs:sequence>
        <xs:element name="creator" type="entityType" minOccurs="0" maxOccurs="unbounded">
          <xs:annotation>
            <xs:documentation> The descriptor creator identifies an 'entity' (a person, group
              of persons or organisation) primarily responsible for creating the content
              of the resource - behind the camera. Different roles may be considered as
              representing a creator, e.g. a producer, an author, etc. Creator is a
              sub-class of Contributor. </xs:documentation>
            </xs:annotation>
          </xs:element>
        <xs:element name="subject" type="subjectType" minOccurs="0" maxOccurs="unbounded">
          <xs:annotation>
            <xs:documentation> The generalised topic that represents the intellectual content
              of the resource. Typically, a subject is expressed by keywords, key phrases.
              Free text, controlled vocabularies, authorities, or formal classification
              schemes (codes) may be employed when selecting descriptive subject terms.

```

Persons as subjects are also placed here. Genre of the content is defined under element "ebucore:type/ebucore:genre" </xs:documentation>

```

</xs:annotation>
</xs:element>
<xs:element name="description" type="descriptionType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation>Free-form text or a narrative to report general notes, abstracts,
      or summaries about the intellectual content of a resource. The information
      may be in the form of a paragraph giving an individual program description,
      anecdotal interpretations, or brief content reviews. The description may
      also consist of outlines, lists, bullet points, edit decision lists,
      indexes, or tables of content, a reference to a graphical representation of
      content or even a pointer (URI, URL) to an external resource. A running
      order can also be provided as a description. For a Radio or television
      programme a running order can be used as description. A description can be
      provided in different languages. </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="publisher" type="entityType" minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation>A publisher is a person, an organization, or a service.
        Typically, the name of a Publisher should be used to indicate the entity
        primarily responsible for distributing or making a resource available to
        others e.g. by broadcasting, selling, leasing, renting and other modes of
        distribution.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="contributor" type="entityType"
      minOccurs="0" maxOccurs="unbounded">
      <xs:annotation>
        <xs:documentation>The descriptor contributor identifies a person or organization
          that has made substantial creative contributions to the content of a
          resource. Refers particularly (but not only) to participation in front of
          the camera. If in doubt whether an entity is a creator or contributor use
          the element contributor.</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="date" type="dateType" minOccurs="0" maxOccurs="unbounded">
        <xs:annotation>
          <xs:documentation>Dates associated with events occurring during the life of the
            resource. Typically, Date will be associated with the creation, modification
            or availability of the resource.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="type" type="typeType" minOccurs="0" maxOccurs="unbounded">
          <xs:annotation>
            <xs:documentation>The nature or genre of the resource. Type includes terms
              describing general categories, functions, genres, or aggregation levels for
              content. Recommended best practice is to select a value from a controlled
              vocabulary. To describe the physical or digital manifestation of the
              resource, use the FORMAT element.</xs:documentation>
            </xs:annotation>
          </xs:element>
          <xs:element name="format" type="formatType" minOccurs="0" maxOccurs="unbounded">
            <xs:annotation>
              <xs:documentation>The physical or digital manifestation of the resource. Use the
                descriptor Format to identify the format of a particular resource as it
                exists in its physical or digital form. Physical form = an actual physical
                form that occupies physical space, e.g. a tape. Digital form = a digital

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file residing on a server or hard drive. Format may be used to determine the software, hardware or other equipment needed to display or operate the resource.

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</xs:annotation>
</xs:element>
<xs:element name="identifier" type="identifierType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A unique, unambiguous reference or identifier for a resource
      within a given context. Best practice is to identify the resource (whether
      analogue or digital) by means of a string or number corresponding to an
      established or formal identification system if one exists. Otherwise, use an
      identification method that is in use within your agency, station, production
      company, office, or institution. It is also possible to enter more than one,
      different but still unique, identifier for the same
      resource.
    </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element ref="dc:source" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation>Reference to the resource (s) from which the current resource is
      derived in whole or in part. If no label or number is available, the title
      and/or the statement of responsibility etc. of the digitized recording is
      recorded here. For a digitized radio programme the production number is
      normally given here. The Recommended best practice is to use a unique
      identifier to identify the physical source that has been used to create the
      digital resource. In the case of a digitized analogue recording, it is the
      recording used for digitization which is the source. For commercial
      recordings the label and number is normally given here. Example: Eurovision
      feed 2007-07-16T19:20:30.45+01:00
    </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="language" type="languageType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> Identifies languages and their use in the intellectual content
      of the resource. Recommended best practice for the values of the Language
      element is defined by RFC 1766, which includes a two-letter Language Code
      (taken from the ISO Standard 639), followed optionally, by a two-letter
      Country Code (taken from the ISO Standard 3166). For example, 'en' for
      English, 'fr' for French, or 'en-uk' for English used in the United Kingdom.
      More contextual information can be provided using the "note"
      attribute.
    </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="relation" type="relationType" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> Recommended best practice is to reference the resource (to which
      the current resource under description is related) by means of a string or
      number conforming to a formal identification system. Relation is used to
      show the relation in content to another resource. For example, "IsPartOf" is
      used to show the relation between a part of a radio programme and the whole
      programme or between a track and a record album. A resource can be
      identified by its title, or an identifier (possibly a URI). The related item
      has its own separate metadata record. Relation is used to provide a name, an
      identification number or ID, or a locator where the related item can be
      found.
    </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="isVersionOf" type="relationType"

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    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to the resource that the current resource is a
        version of </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="hasVersion" type="relationType"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to another version of the resource
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="isReplacedBy" type="relationType"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to a resource replacing the current resource
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="replaces" type="relationType"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to a resource that the current resource replaces
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="isRequiredBy" type="relationType"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to a resource requiring the current resource
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="requires" type="relationType" minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to a resource that the current resource requires
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="isPartOf" type="relationType" minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to a resource that the current resource is a part of
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="hasPart" type="relationType" minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to a resource that forms part of the current
        resource </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="isReferencedBy" type="relationType"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> A reference to a resource that references the current resource
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="references" type="relationType"

```



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    minOccurs="0" maxOccurs="unbounded">
<xs:annotation>
  <xs:documentation> A reference to a resource that the current resource references
</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="isFormatOf" type="relationType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A reference to a resource with which the current resource shares
      a format </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="hasFormat" type="relationType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A format in which the resource is also available
  </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="isEpisodeOf" type="relationType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A reference to a series the current resource is an episode of
  </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="isMemberOf" type="relationType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A reference to a group the current resource is an member of
  </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="coverage" type="coverageType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> Coverage is used to show various time and place aspects of the
      subject of the content. Coverage will typically include spatial location (a
      place name or geographic coordinates), temporal period (a period label,
      date, or date range) or jurisdiction (such as a named administrative
      entity). Recommended best practice is to select a value from a controlled
      vocabulary (for example, the Thesaurus of Geographic Names) and that, where
      appropriate, named places or time periods be used in preference to numeric
      identifiers such as sets of coordinates or date ranges.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="rights" type="rightsType" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> An all-purpose field to identify information (rights management
      statement or reference to a service providing such information e.g. via a
      URL) about copyright, intellectual property rights or other property rights
      held in and over a resource, stating whether access is open or restricted in
      some way. If dates, times, territories and availability periods are
      associated with a right, they should be included. If the Rights element is
      absent, no assumptions can be made about the status of these and other
      rights with respect to the resource.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="version" type="dc:elementType" minOccurs="0">

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<xs:annotation>
  <xs:documentation xml:lang="en">UK Version, US Version, home video version, etc.
    Mapping to Dublin Core would be made using a description
    element</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="publicationHistory" type="publicationHistoryType" minOccurs="0">
  <xs:annotation>
    <xs:documentation> To provide information on the publication history.
    </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="part" type="coreMetadataType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation>To identify parts/segments/fragments within the
    resource.</xs:documentation>
  </xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="titleType">
  <xs:annotation>
    <xs:documentation> The name given to a resource
    e.g. a media item, media object, sequence.
    For a series – use the series title; for a programme – a programme title; for an
    item – an item title. etc. Titles are recorded as they appear.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="dc:title">
      <xs:annotation>
        <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
        Core. Free-text to provide the main title by which the resource is known.
        The title can be provided in different languages. The language in which the
        title is provided can be provided using elementType's lang attribute.
        Example: 'the fifth element' </xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
  <xs:attribute name="attributiondate" type="xs:date">
    <xs:annotation>
      <xs:documentation> Defines the date of attribution of this Title. </xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="note" type="xs:string">
    <xs:annotation>
      <xs:documentation> Optional additional contextual information. </xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
<xs:complexType name="alternativeTitleType">
  <xs:annotation>
    <xs:documentation> The name given to a resource
    e.g. a media item, media object, sequence.
    For a series – use the series title; for a programme – a programme title; for an
    item – an item title. etc. Titles are recorded as they appear.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="dc:title">
      <xs:annotation>

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<xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
Core. Free-text to provide alternative titles by which the resource is
known. The language in which the title is provided can be provided using
elementType's lang attribute. Example: 'the fifth element' </xs:documentation>
</xs:annotation>
</xs:element>
</xs:sequence>
<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation> The typeGroup is used to define the type of alternative title. This
    can be an associated title like a series title.</xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attributeGroup ref="statusGroup">
  <xs:annotation>
    <xs:documentation> The statusGroup is used to define the status of the Title such as
    short, long, full, abridged, working, transmission, published, international,
    subtitle, original, secondary, alternative, pledged, etc. The name of the format
    can be provided in the form of a text label, or a link to a code of a
    classification scheme, optionally accompanied by a definition. the status 'main'
    shall not be used for alternativeTitle as this applies to the Title
    only.</xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attributeGroup ref="dateGroup">
  <xs:annotation>
    <xs:documentation> Defines the date of attribution of this Title. </xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attribute name="note" type="xs:string">
  <xs:annotation>
    <xs:documentation> Optional additional contextual information. </xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
<xs:complexType name="identifierType">
  <xs:annotation>
    <xs:documentation> A unique, unambiguous reference or identifier for a resource within a
    given context. Best practice is to identify the resource (whether analogue or
    digital) by means of a string or number corresponding to an established or formal
    identification system if one exists. Otherwise, use an identification method that is
    in use within your agency, station, production company, office, or institution. It
    is also possible to enter different but unique identifiers for the same resource.
  </xs:documentation>
  </xs:annotation>
</xs:sequence>
  <xs:element ref="dc:identifier">
    <xs:annotation>
      <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
      Core. Free text to provide an identifier. Example: 06.0A.2B.34.01.01.01.01
    </xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
  <xs:attribute name="attributor" type="xs:string">
    <xs:annotation>
      <xs:documentation> To identify the source of attribution of the
      identifier.</xs:documentation>
    </xs:annotation>
  </xs:attribute>

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<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation> The typeGroup is used to define the type of Identifier e.g. Main or
      Alternative </xs:documentation>
    </xs:annotation>
  </xs:attributeGroup>
<xs:attributeGroup ref="formatGroup">
  <xs:annotation>
    <xs:documentation> Use to define the format and possibly syntax of the identifier. Used
      in combination with the resource Identifier. It can denote the agency or
      institution which specified or assigned it e.g. SMPTE UMID, ISO ISAN, IETF URI,
      ISRC, custom.</xs:documentation>
    </xs:annotation>
  </xs:attributeGroup>
<xs:attribute name="note" type="xs:string">
  <xs:annotation>
    <xs:documentation> Optional additional contextual information. </xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
<xs:complexType name="subjectType">
  <xs:annotation>
    <xs:documentation>The generalised topic of that represents the intellectual content of the
      resource. Typically, a subject is expressed by keywords, key phrases, or even
      specific classification codes. Controlled vocabularies, authorities, or formal
      classification schemes may be employed when selecting descriptive subject terms. It
      is possible to employ both keywords, derived from a formal classification scheme,
      such as Dewey or UDC, and genres/subgenres such as those produced by TV-Anytime or
      Escort, to cover Subject(s) and Genre(s) and enter as appropriate Subject Type
      below. Persons as subjects are also placed here. Genre of the content is placed
      under element Type. </xs:documentation>
    </xs:annotation>
  <xs:sequence>
    <xs:element ref="dc:subject">
      <xs:annotation>
        <xs:documentation>To express the subject in the form of free text. </xs:documentation>
        </xs:annotation>
      </xs:element>
    <xs:element name="subjectCode" type="xs:anyURI" minOccurs="0">
      <xs:annotation>
        <xs:documentation>To alternatively express the subject using predefined terms
          expressed by classification codes. Reference data: - Library of Congress
          Subject Heading (LCSH), Library of Congress Classification (LCC), Medical
          Subject Headings (MeSH), Dewey Decimal Classification (DDC), Dansk
          decimalklassedeling 5.utgave (DK5), Klassifikasjonssystem för svenska
          bibliotek (SAB), Universal Decimal Classification (UDC), Norske emneord -
          http://cv.ipc.org/newscodes/subjectcode/. Example:
          http://cv.ipc.org/newscodes/subjectcode/#15065000 </xs:documentation>
        </xs:annotation>
      </xs:element>
    <xs:element name="subjectDefinition" type="xs:string" minOccurs="0">
      <xs:annotation>
        <xs:documentation>An optional definition. Example: 'the subject is about tennis
          (sport, game)' </xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation>To define the source of reference for subject such as a reference

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    document or classification scheme.</xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attribute name="note" type="xs:string">
  <xs:annotation>
    <xs:documentation> Optional additional contextual information. </xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
<xs:complexType name="typeType">
  <xs:annotation>
    <xs:documentation>The nature or genre of the content of the resource. Type includes terms
    describing general categories, functions, genres, or aggregation levels for content.
    Recommended best practice is to select a value from a controlled vocabulary. To
    describe the physical or digital manifestation of the resource, use the FORMAT
    element. </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="dc:type" minOccurs="0" maxOccurs="unbounded">
      <xs:annotation>
        <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
        Core. </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="genre" minOccurs="0" maxOccurs="unbounded">
      <xs:annotation>
        <xs:documentation>A type element specifically dedicated to the description of a
        resource genre</xs:documentation>
      </xs:annotation>
      <xs:complexType>
        <xs:attributeGroup ref="typeGroup">
          <xs:annotation>
            <xs:documentation>To define the Type reference data.</xs:documentation>
          </xs:annotation>
        </xs:attributeGroup>
      </xs:complexType>
    </xs:element>
    <xs:element name="objectType" minOccurs="0" maxOccurs="unbounded">
      <xs:annotation>
        <xs:documentation>A type element specifically dedicated to the description of type
        of resource being describe e.g. programme, item, episode</xs:documentation>
      </xs:annotation>
      <xs:complexType>
        <xs:attributeGroup ref="typeGroup">
          <xs:annotation>
            <xs:documentation>To define the Type reference data.</xs:documentation>
          </xs:annotation>
        </xs:attributeGroup>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
  <xs:attribute name="note" type="xs:string">
    <xs:annotation>
      <xs:documentation> Optional additional contextual information. </xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
<xs:complexType name="descriptionType">
  <xs:annotation>
    <xs:documentation>Free-form text or a narrative to report general notes, abstracts, or

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summaries about the intellectual content of a resource. The information may be in the form of a paragraph giving an individual program description, anecdotal interpretations, or brief content reviews. The description may also consist of outlines, lists, bullet points, edit decision lists, indexes, or tables of content, a reference to a graphical representation of content or even a pointer (URI, URL) to an external resource. For a Radio or television programme a running order can be used as description. A description can be provided in different languages.

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</xs:documentation>
</xs:annotation>
<xs:sequence>
  <xs:element ref="dc:description">
    <xs:annotation>
      <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
        Core. Free text to provide a description of the resource. The description
        can be repeated in different languages as specified by the entityType's lang
        attribute. The type of description is defined in the type group of
        attributes.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation> To define the form of presentation for the information: Annotation,
      abstract, summary, review, table of content, synopsis, shot list, edit decision
      list, promotional information, purpose, script, outline, rundown,
      selection/excerpt, transcript, bookmarks, theme, highlights, running order, etc.
    </xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attribute name="note" type="xs:string">
  <xs:annotation>
    <xs:documentation> Optional additional contextual information. </xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
<xs:complexType name="coverageType">
  <xs:annotation>
    <xs:documentation> Coverage will typically include spatial location (a place name or
      geographic coordinates), temporal period (a period label, date, or date range) or
      jurisdiction (such as a named administrative entity). Recommended best practice is
      to select a value from a controlled vocabulary (for example, the Thesaurus of
      Geographic Names) and that, where appropriate, named places or time periods be used
      in preference to numeric identifiers such as sets of coordinates or date
      ranges.</xs:documentation>
  </xs:annotation>
<xs:sequence>
  <xs:element ref="dc:coverage" minOccurs="0">
    <xs:annotation>
      <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
        Core. </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="temporal" minOccurs="0">
    <xs:annotation>
      <xs:documentation> Temporal characteristics of the content of the resource. To
        indicate e.g. specific date, time or period aspects of the subject of the
        resource in complement to Description. </xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>

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<xs:element name="PeriodOfTime" minOccurs="0">
  <xs:complexType>
    <xs:attributeGroup ref="dateGroup">
      <xs:annotation>
        <xs:documentation>The period of time depicted in the
          resource.</xs:documentation>
      </xs:annotation>
    </xs:attributeGroup>
  </xs:complexType>
</xs:element>
</xs:sequence>
<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation>To precise the type of temporal information
      provided.</xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attribute name="note" type="xs:string">
  <xs:annotation>
    <xs:documentation>Optional additional contextual information.
    </xs:documentation>
  </xs:annotation>
</xs:attribute>
<xs:attribute name="periodId" type="xs:anyURI">
  <xs:annotation>
    <xs:documentation>An identifier to support the management of time periods
      (e.g. historical or repetitive event) in databases and
      RDF</xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
</xs:element>
<xs:element name="spatial" minOccurs="0">
  <xs:annotation>
    <xs:documentation>Spatial characteristics of the content of the resource. To
      indicate e.g. specific place and location aspects of the subject of the
      resource in complement to Description. </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element name="location" minOccurs="0" maxOccurs="unbounded">
        <xs:annotation>
          <xs:documentation>To indicate e.g. specific place and location aspects
            of the resource in complement to Description.</xs:documentation>
        </xs:annotation>
        <xs:complexType>
          <xs:sequence>
            <xs:element name="name" type="xs:string" minOccurs="0">
              <xs:annotation>
                <xs:documentation>Any location name in free
                  text</xs:documentation>
              </xs:annotation>
            </xs:element>
            <xs:sequence minOccurs="0">
              <xs:annotation>
                <xs:documentation>Optional geospatial coordinates. 'posy'
                  is the latitude. 'posx' is the longitude. Both are
                  expressed in digital degrees</xs:documentation>
              </xs:annotation>
              <xs:element name="posy" type="xs:float"/>

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    <xs:element name="posx" type="xs:float"/>
  </xs:sequence>
  <xs:element name="code" type="xs:anyURI" minOccurs="0">
    <xs:annotation>
      <xs:documentation>A location identified by a code from a
        predefined list of locations.</xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation>To precise the type of place and
      location.</xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attribute name="note" type="xs:string">
  <xs:annotation>
    <xs:documentation>To provide additional information on the type
      of location described, e.g. countries, regions,
      cities</xs:documentation>
  </xs:annotation>
</xs:attribute>
<xs:attribute name="locationId" type="xs:anyURI">
  <xs:annotation>
    <xs:documentation>An identifier to support the management of
      location in databases and RDF</xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="rightsType">
  <xs:annotation>
    <xs:documentation>An all-purpose field to identify information (rights management statement
      or reference to a service providing such information e.g. via a URL) about
      copyright, intellectual property rights or other property rights held in and over a
      resource, stating whether open access or restricted in some way. If dates, times,
      territories and availability periods are associated with a right, they should be
      included. If the Rights element is absent, no assumptions can be made about the
      status of these and other rights with respect to the resource. </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="dc:rights" minOccurs="0">
      <xs:annotation>
        <xs:documentation>The EBU core metadata set is built as a refinement of the Dublin
          Core. An element to express any form of rights related
          matters.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="rightsLink" type="xs:anyURI" minOccurs="0">
      <xs:annotation>
        <xs:documentation>A url pointing to a declaration of rights</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="rightsHolder" type="entityType" minOccurs="0">
      <xs:annotation>

```

```

    <xs:documentation>To identify the person or organisation holding or managing the
    rights related to the resource. </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="exploitationIssues" type="dc:elementType" minOccurs="0">
  <xs:annotation>
    <xs:documentation>Use to state any other restrictions, such as non-rights ones,
    e.g. legal. State by media, territory, scope (restriction on whole item or
    extracts) and possibly language. The presence of this information can be
    used by asset management system implementing traffic lights like mechanism
    to signal that content may be subject to particular restrictions to be
    clarified before exploitation. </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="coverage" type="coverageType" minOccurs="0">
  <xs:annotation>
    <xs:documentation> Specifies a specific start date, end date or period for the
    availability of the item or the date from which the rights or exploitation
    issues apply. It may refer to start dates for the availability of an item
    that is used within a particular geographical area e.g. broadcast locally,
    regionally, nationally or internationally, or for web-based distribution. A
    specific time may also be associated with the date.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="rightsClearanceFlag" type="xs:boolean" minOccurs="0">
  <xs:annotation>
    <xs:documentation> A flag to signal if content is subject to rights open issues
  </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="disclaimer" type="dc:elementType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A field for a disclaimer about the content, its content, and its
    use. </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="rightsId" type="identifierType"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A identifier related to rights, e.g. attributed for a particular
    purpose by a specific agency in the context of use and exploitation.
  </xs:documentation>
  </xs:annotation>
</xs:element>
</xs:sequence>
<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation> To define the type of rights information provided.</xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
<xs:attribute name="note" type="xs:string">
  <xs:annotation>
    <xs:documentation> Optional additional contextual information. </xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
<xs:complexType name="formatType">
  <xs:annotation>
    <xs:documentation>The physical or digital manifestation of the resource. Use the descriptor

```

Format to identify the format of a particular resource as it exists in its physical or digital form. Physical form = an actual physical form that occupies physical space, e.g. a tape. Digital form = a digital file residing on a server or hard drive. Format may be used to determine the software, hardware or other equipment needed to display or operate the resource. </xs:documentation>

```

</xs:annotation>
<xs:sequence>
  <xs:element ref="dc:format" minOccurs="0">
    <xs:annotation>
      <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
        Core. Free text to provide information on the format.</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="width" minOccurs="0">
    <xs:annotation>
      <xs:documentation>The width of the image or picture. Used as numerator to define
        the aspect ratio for video content. Otherwsie, see aspect
        ratio.</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:simpleContent>
        <xs:extension base="xs:nonNegativeInteger">
          <xs:attribute name="unit" type="xs:string">
            <xs:annotation>
              <xs:documentation> An attribute to specify the unit in which the
                width is expressed. </xs:documentation>
            </xs:annotation>
          </xs:attribute>
        </xs:extension>
      </xs:simpleContent>
    </xs:complexType>
  </xs:element>
  <xs:element name="height" minOccurs="0">
    <xs:annotation>
      <xs:documentation>The height of the image or picture. Used as denominator to define
        the aspect ratio for video content. Otherwsie, see aspect
        ratio.</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:simpleContent>
        <xs:extension base="xs:nonNegativeInteger">
          <xs:attribute name="unit" type="xs:string">
            <xs:annotation>
              <xs:documentation> An attribute to specify the unit in which the
                height is expressed. </xs:documentation>
            </xs:annotation>
          </xs:attribute>
        </xs:extension>
      </xs:simpleContent>
    </xs:complexType>
  </xs:element>
  <xs:element name="medium" minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation>The material or physical carrier of the resource. If a file, it
        should be the carrier format.</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attributeGroup ref="typeGroup"/>
    </xs:complexType>
  </xs:element>

```



```

<xs:element name="mimeType" minOccurs="0">
  <xs:annotation>
    <xs:documentation>Define the main MIME type as defined by IANA: e.g. audio, video,
      text, application</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attributeGroup ref="typeGroup"/>
    </xs:complexType>
  </xs:element>
<xs:element name="videoFormat" minOccurs="0">
  <xs:annotation>
    <xs:documentation> used to list all the characteristics of the video signal
    </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element name="aspectRatio" minOccurs="0">
        <xs:annotation>
          <xs:documentation>A string to define e.g. the ratio of the picture (the
            width by the height), for instance '4:3' or '16 9' (rational).
            The format of the aspect ratio is precised in the format
            attributes</xs:documentation>
          </xs:annotation>
          <xs:complexType>
            <xs:simpleContent>
              <xs:extension base="xs:string">
                <xs:attributeGroup ref="formatGroup">
                  <xs:annotation>
                    <xs:documentation>To define the type of format used to
                      represent the aspect ratio and an example in the
                      definition e.g. formatLabel='rational' and
                      formatDefinition='e.g. 16 9 (number, space,
                      number)'</xs:documentation>
                  </xs:annotation>
                  </xs:attributeGroup>
                  <xs:attribute name="note" type="xs:string"/>
                </xs:extension>
              </xs:simpleContent>
            </xs:complexType>
          </xs:element>
        <xs:element name="videoEncoding" minOccurs="0">
          <xs:annotation>
            <xs:documentation> Used to express the encoding parameters of the
              resource e.g. H264 for a video channel.</xs:documentation>
          </xs:annotation>
          <xs:complexType>
            <xs:attributeGroup ref="typeGroup"/>
          </xs:complexType>
        </xs:element>
      <xs:element name="videoTrack" minOccurs="0" maxOccurs="unbounded">
        <xs:annotation>
          <xs:documentation>To describe the main features of video tracks such as
            in mutliview systems</xs:documentation>
          </xs:annotation>
          <xs:complexType>
            <xs:sequence>
              <xs:element name="videoTrackId" type="xs:NMTOKEN" minOccurs="0">
                <xs:annotation>
                  <xs:documentation> The track ID or track number
                  </xs:documentation>

```



```

    </xs:annotation>
  </xs:element>
  <xs:element name="videoTrackName" type="xs:string" minOccurs="0">
    <xs:annotation>
      <xs:documentation> The track name </xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
<xs:attributeGroup ref="typeGroup">
  <xs:annotation>
    <xs:documentation> The type of video track e.g. particular view
      angle. </xs:documentation>
  </xs:annotation>
</xs:attributeGroup>
</xs:complexType>
</xs:element>
<xs:element name="videoTechnicalAttributeString"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A user defined technical attribute to provide
      information on the Video Format (in addition to the video
      encoding format already provided in channel/encoding): colour
      scheme, scanning format, etc. Examples: 1/ value=50,
      Black&White, value=progressive attribute label=scan
      type, etc.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:string">
        <xs:attributeGroup ref="typeGroup"/>
        <xs:attributeGroup ref="formatGroup"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
<xs:element name="videoTechnicalAttributeInteger"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A user defined technical attribute to provide
      information on the Video Format (in addition to the video
      encoding format already provided in channel/encoding): frame
      rate, sampling rate, etc. Examples: frame rate value=50,
      attribute label= frame rate , the bitrate for this particular
      channel, etc.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:integer">
        <xs:attributeGroup ref="typeGroup"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
<xs:element name="videoTechnicalAttributeFlag"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A user defined technical attribute to provide
      information on the Video Format (in addition to the video
      encoding format already provided in channel/encoding): e.g. HD
      flag</xs:documentation>

```



```

</xs:annotation>
<xs:complexType>
  <xs:simpleContent>
    <xs:extension base="xs:boolean">
      <xs:attributeGroup ref="typeGroup"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>
</xs:element>
<xs:element name="videoTechnicalAttributeFloat"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> A user defined technical attribute to provide
      information on the Video Format (in addition to the video
      encoding format already provided in
      channel/encoding)</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:float">
        <xs:attributeGroup ref="typeGroup"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="audioFormat" minOccurs="0">
  <xs:annotation>
    <xs:documentation> used to list all the characteristics of the audio signal
  </xs:documentation>
</xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element name="audioEncoding" minOccurs="0">
        <xs:annotation>
          <xs:documentation> To define the audio compression format of the
            resource e.g. AAC for an audio channel.</xs:documentation>
        </xs:annotation>
        <xs:complexType>
          <xs:attributeGroup ref="typeGroup"/>
        </xs:complexType>
      </xs:element>
      <xs:element name="audioTrackConfiguration" minOccurs="0">
        <xs:annotation>
          <xs:documentation> To define the audio track configuration.
            Used to express the arrangement or audio tracks e.g.
            'stereo', '2+1', 'surround', 'surround (7+1)'</xs:documentation>
        </xs:annotation>
        <xs:complexType>
          <xs:simpleContent>
            <xs:extension base="xs:string">
              <xs:attributeGroup ref="typeGroup"/>
            </xs:extension>
          </xs:simpleContent>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:annotation>
    <xs:documentation>

```



```

<xs:documentation> To describe the track allocation e.g. in conformance with
EBU R123 </xs:documentation>
</xs:annotation>
<xs:complexType>
  <xs:sequence>
    <xs:element name="audioTrackId" type="xs:NMTOKEN" minOccurs="0">
      <xs:annotation>
        <xs:documentation> The track number </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="audioTrackName" type="xs:string" minOccurs="0">
      <xs:annotation>
        <xs:documentation> The track name</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="audioTrackLanguage" type="languageType"
      minOccurs="0">
      <xs:annotation>
        <xs:documentation> The track language </xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
  <xs:attributeGroup ref="typeGroup">
    <xs:annotation>
      <xs:documentation> The track type </xs:documentation>
    </xs:annotation>
  </xs:attributeGroup>
</xs:complexType>
</xs:element>
<xs:element name="audioTechnicalAttributeString"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> To provide information on the Audio Format(in
    addition to the audio encoding format already provided in
    channel/encoding): restricted for information that cannot be
    represented in the format of integers or boolean
    flags</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:string">
        <xs:attributeGroup ref="typeGroup"/>
        <xs:attributeGroup ref="formatGroup"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
<xs:element name="audioTechnicalAttributeInteger"
  minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> To provide information on the Audio Format(in
    addition to the audio encoding format already provided in
    channel/encoding): sample rate, etc. Examples: 1/ value=44800,
    attribute label = sample rate, attribute definition=Hz, the
    bitrate for this particular channel, etc. </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:integer">
        <xs:attributeGroup ref="typeGroup"/>

```



```

        </xs:extension>
        </xs:simpleContent>
    </xs:complexType>
</xs:element>
<xs:element name="audioTechnicalAttributeFlag"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
        <xs:documentation> To provide information on the Audio Format(in
            addition to the audio encoding format already provided in
            channel/encoding) in the form of flags e.g. constant bitrate
            flag </xs:documentation>
    </xs:annotation>
    <xs:complexType>
        <xs:simpleContent>
            <xs:extension base="xs:boolean">
                <xs:attributeGroup ref="typeGroup"/>
            </xs:extension>
        </xs:simpleContent>
    </xs:complexType>
</xs:element>
<xs:element name="audioTechnicalAttributeFloat"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
        <xs:documentation/>
    </xs:annotation>
    <xs:complexType>
        <xs:simpleContent>
            <xs:extension base="xs:float">
                <xs:attributeGroup ref="typeGroup"/>
            </xs:extension>
        </xs:simpleContent>
    </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="fileFormat" minOccurs="0">
    <xs:annotation>
        <xs:documentation> To provide information on the file / wrapper format in
            complement to the stream encoding information provided in 'channel', e.g.
            mp3, wave, Quicktime, ogg..</xs:documentation>
    </xs:annotation>
    <xs:complexType>
        <xs:attributeGroup ref="formatGroup"/>
    </xs:complexType>
</xs:element>
<xs:element name="captioningFormat" minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
        <xs:documentation> Used to provide information on the captioning format and purpose
    </xs:documentation>
    </xs:annotation>
    <xs:complexType>
        <xs:complexContent>
            <xs:extension base="languageType">
                <xs:attributeGroup ref="formatGroup">
                    <xs:annotation>
                        <xs:documentation>To specify the format of captioning / subtitling
                            being used</xs:documentation>
                    </xs:annotation>
                </xs:attributeGroup>
            </xs:extension>
        </xs:complexContent>
    </xs:complexType>

```



```

</xs:attributeGroup>
<xs:attribute name="captioningSourceUri" type="xs:anyURI">
  <xs:annotation>
    <xs:documentation>A pointer to the file with the captioning /
      subtitling.</xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:extension>
</xs:complexContent>
</xs:complexType>
</xs:element>
<xs:element name="signingFormat" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation>Used to provide information on the signing format and purpose
    </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="languageType">
        <xs:attributeGroup ref="formatGroup"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
<xs:element name="start" minOccurs="0">
  <xs:annotation>
    <xs:documentation>The beginning point for playback of a time-based media item, such
      as digital video or audio. Use in combination with Duration to identify a
      sequence or segment of a media item that has a fixed start time and end
      time.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:choice>
      <xs:element name="timecode">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[0-9][0-9]:[0-5][0-9]:[0-5][0-9]:[0-9][0-9]" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="normalPlayTime" type="xs:time"/>
      <xs:element name="editUnitNumber" type="editUnitNumberType">
        <xs:annotation>
          <xs:documentation>A number of edit units. An editUnit is the inverse of
            the edit rate, or corrected edit rate as the result of
            editUnit=1/(editrate*(factorNumerator/factorDenominator))
          </xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="time">
        <xs:complexType>
          <xs:simpleContent>
            <xs:extension base="xs:string">
              <xs:attributeGroup ref="formatGroup"/>
            </xs:extension>
          </xs:simpleContent>
        </xs:complexType>
      </xs:element>
    </xs:choice>
  </xs:complexType>

```



```

</xs:element>
<xs:element name="duration" minOccurs="0">
  <xs:annotation>
    <xs:documentation>The time duration/extent of the resource</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:choice>
      <xs:element name="timecode">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[0-9][0-9]:[0-5][0-9]:[0-5][0-9]:[0-9][0-9]" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="normalPlayTime" type="xs:duration"/>
      <xs:element name="editUnitNumber" type="editUnitNumberType">
        <xs:annotation>
          <xs:documentation>A number of edit units. An editUnit is the inverse of
            the edit rate, or corrected edit rate as the result of
            editUnit=1/(editrate*(factorNumerator/factorDenominator))
          </xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="time">
        <xs:complexType>
          <xs:simpleContent>
            <xs:extension base="xs:string">
              <xs:attributeGroup ref="formatGroup"/>
            </xs:extension>
          </xs:simpleContent>
        </xs:complexType>
      </xs:element>
    </xs:choice>
  </xs:complexType>
</xs:element>
<xs:element name="fileSize" type="xs:nonNegativeInteger" minOccurs="0">
  <xs:annotation>
    <xs:documentation>To indicate the storage requirements or file size of a digital
      resource. The file size is expressed in bytes.</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="filename" type="xs:string" minOccurs="0">
  <xs:annotation>
    <xs:documentation> The name of the file as it appears in the location path or url.
  </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="locator" type="xs:string" minOccurs="0">
  <xs:annotation>
    <xs:documentation>An "address for a resource". For an organisation or producer
      acting as caretaker for a media resource, Format Location may contain
      information about a specific shelf location for an asset, including an
      organisation's name, departmental name, shelf id. and contact information.
      The Format Location for a data file or web page may include a complete URI
      with a domain, path, filename or html URL. Examples: "Archives Building A,
      Row J, Shelf 2", "d://playout/server/content.mpg",
      "http://www.ebu.ch/CorporateVideo.avi".</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="technicalAttributeString" minOccurs="0" maxOccurs="unbounded">

```

```

<xs:annotation>
  <xs:documentation> To provide a user defined technical attribute. </xs:documentation>
</xs:annotation>
<xs:complexType>
  <xs:simpleContent>
    <xs:extension base="xs:string">
      <xs:attributeGroup ref="typeGroup"/>
      <xs:attributeGroup ref="formatGroup"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>
</xs:element>
<xs:element name="technicalAttributeInteger" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> To provide a user defined technical attribute. </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:integer">
        <xs:attributeGroup ref="typeGroup"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
<xs:element name="technicalAttributeFlag" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> To provide a user defined technical attribute. </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:boolean">
        <xs:attributeGroup ref="typeGroup"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
<xs:element name="technicalAttributeFloat" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation> To provide a user defined technical attribute. </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:simpleContent>
      <xs:extension base="xs:float">
        <xs:attributeGroup ref="typeGroup"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="editUnitNumberType">
  <xs:simpleContent>
    <xs:extension base="xs:long">
      <xs:attribute name="editRate" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>The base number of frames or samples
            per seconds. This base number can be corrected by
            a factor calculated as the result of
            'factorNumerator/factorDenominator'
          </xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>

```



```

    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="factorNumerator" type="xs:positiveInteger" default="1">
    <xs:annotation>
      <xs:documentation>The numerator of the correction factor</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="factorDenominator" type="xs:positiveInteger" default="1">
    <xs:annotation>
      <xs:documentation>The denominator of the correction factor </xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:extension>
</xs:simpleContent>
</xs:complexType>
<xs:complexType name="relationType">
  <xs:annotation>
    <xs:documentation>Recommended best practice is to reference the resource by means of a
      string or number conforming to a formal identification system. Relation is used to
      show the relation in content to another resource. For example, "IsPartOf" is used to
      show the relation between a part of a radio programme and the whole programme, or
      between a track and a record album. A resource can be identified by its title, or
      preferably by an identifier. Relation is used to provide a name, locator, accession,
      identification number or ID where the related item can be obtained or found.
    </xs:documentation>
  </xs:annotation>
  <xs:choice>
    <xs:element ref="dc:relation">
      <xs:annotation>
        <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
          Core. A title would be given using this element.</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="relationIdentifier" type="identifierType">
      <xs:annotation>
        <xs:documentation> An identifier would be given using this element.
      </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="relationLink" type="xs:anyURI">
      <xs:annotation>
        <xs:documentation> A link to related material. </xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:choice>
  <xs:attributeGroup ref="typeGroup">
    <xs:annotation>
      <xs:documentation>To show the type of relation to another resource, e.g. identifies
        ways in which the resource is related by intellectual content to some other
        resource.</xs:documentation>
    </xs:annotation>
  </xs:attributeGroup>
  <xs:attribute name="runningOrderNumber" type="xs:integer">
    <xs:annotation>
      <xs:documentation> If exists, it provides the ranking/running order within an ordered
        list.</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="note" type="xs:string">
    <xs:annotation>

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    <xs:documentation> Optional additional contextual information. </xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
<xs:complexType name="languageType">
  <xs:annotation>
    <xs:documentation> MODIF VERSION 1.2</xs:documentation>
    <xs:documentation>Identifies languages and their use in the intellectual content of the
      resource. Recommended best practice for the values of the Language element is
      defined by RFC 1766, which includes a two-letter Language Code (taken from the ISO
      Standard 639), followed optionally, by a two-letter Country Code (taken from the ISO
      Standard 3166). For example, 'en' for English, 'fr' for French, or 'en-UK' for
      English used in the United Kingdom. The usage of the language is also defined.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="dc:language" minOccurs="0">
      <xs:annotation>
        <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
          Core. </xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>
  <xs:attributeGroup ref="typeGroup">
    <xs:annotation>
      <xs:documentation> Indicates the purpose of the language described by the Language
        element e.g. Main original language, main dubbed language, additional original
        language, additional dubbed language, descriptive video information,
        supplemental commentary, Director's commentary, audio description, supplementary
        audio programme, educational notes, voice over, original commentary, dubbed
        commentary, original narration, dubbed narration, dubbed dialogue, interviewer
        language, interviewee language, text description for the hard-of-hearing,
        titles, subtitles, song lyrics, sign language, dubbed sign language, transcript,
        caption, open caption, closed caption.</xs:documentation>
    </xs:annotation>
  </xs:attributeGroup>
  <xs:attribute name="note" type="xs:string">
    <xs:annotation>
      <xs:documentation> Optional additional contextual information. </xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
<xs:complexType name="dateType">
  <xs:annotation>
    <xs:documentation>Dates associated with events occurring during the life of the resource.
      Typically, Date will be associated e.g. with the creation or availability of the
      resource. </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="dc:date" minOccurs="0" maxOccurs="unbounded">
      <xs:annotation>
        <xs:documentation> The EBU core metadata set is built as a refinement of the Dublin
          Core. </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="created" minOccurs="0">
      <xs:annotation>
        <xs:documentation>To specify the creation date for a particular version or
          rendition of a resource across its life cycle. It is the moment in time that
          the resource was finalized during its production process and is forwarded to

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other divisions or agencies to make it ready for publication or distribution. A specific time may also be associated with the date. </xs:documentation>

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</xs:annotation>
<xs:complexType>
  <xs:attributeGroup ref="dateGroup"/>
</xs:complexType>
</xs:element>
<xs:element name="issued" minOccurs="0">
  <xs:annotation>
    <xs:documentation>Date of formal issuance (e.g. publication) of the resource.
    Specifies the formal date for a particular version or rendition of a
    resource has been made ready or officially released for distribution,
    publication or consumption, e.g. the broadcasting date of a radio programme.
    A specific time may also be associated with the date. </xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attributeGroup ref="dateGroup"/>
  </xs:complexType>
</xs:element>
<xs:element name="modified" minOccurs="0">
  <xs:annotation>
    <xs:documentation>The date when the resource was last modified</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attributeGroup ref="dateGroup"/>
  </xs:complexType>
</xs:element>
<xs:element name="digitised" minOccurs="0">
  <xs:annotation>
    <xs:documentation>The date when the resource was digitised</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attributeGroup ref="dateGroup"/>
  </xs:complexType>
</xs:element>
<xs:element name="alternative" minOccurs="0" maxOccurs="unbounded">
  <xs:annotation>
    <xs:documentation>An alternative particular date for which the type can be
    defined.</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attributeGroup ref="dateGroup"/>
    <xs:attributeGroup ref="typeGroup"/>
  </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:complexType name="publicationHistoryType">
  <xs:annotation>
    <xs:documentation>To provide information about the publication history which falls outside
    the entries under transmission date/times below. </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="firstTransmissionDate" type="xs:date" minOccurs="0">
      <xs:annotation>
        <xs:documentation>The first transmission date</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="firstTransmissionTime" type="xs:time" minOccurs="0">

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<xs:annotation>
  <xs:documentation>The first transmission time</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="firstTransmissionChannel" type="xs:string" minOccurs="0">
  <xs:annotation>
    <xs:documentation>The channel on which the title was first transmitted
  </xs:documentation>
  </xs:annotation>
</xs:element>
<xs:sequence minOccurs="0" maxOccurs="unbounded">
  <xs:element name="repeatDate" type="xs:date" minOccurs="0">
    <xs:annotation>
      <xs:documentation>The date on which content was re-transmitted
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="repeatTime" type="xs:time" minOccurs="0">
    <xs:annotation>
      <xs:documentation>The time on which content was re-transmitted
    </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="repeatChannel" type="dc:elementType" minOccurs="0">
    <xs:annotation>
      <xs:documentation>The channel on which the resource was re-transmitted
    </xs:documentation>
    </xs:annotation>
  </xs:element>
</xs:sequence>
</xs:sequence>
</xs:complexType>
<xs:complexType name="entityType">
  <xs:annotation>
    <xs:documentation>To identify a person, group of persons or organisation
  </xs:documentation>
</xs:annotation>
<xs:sequence>
  <xs:element name="contactDetails" type="contactDetailsType"
    minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> Minimum information providing means to further identify and
        contact the entity. </xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="organisationDetails" type="organisationDetailsType" minOccurs="0">
    <xs:annotation>
      <xs:documentation>Minimum information providing means to further identify and
        contact the entity as an organisation. Cardinality is '1'. Only one
        organisation is acting as an entity or only one organisation is associated
        to a person in relation to is occupation in the context of the current
        content description</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="role" minOccurs="0" maxOccurs="unbounded">
    <xs:annotation>
      <xs:documentation> Used to identify the function fulfilled by the person, group or
        organisation described as an entity. This is used to detail the role of a
        'contributor'. This also applies to e.g. 'creator' as several functions can
        be seen as participating to the creative process </xs:documentation>
    </xs:annotation>
  </xs:element>

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</xs:annotation>
<xs:complexType>
  <xs:attributeGroup ref="typeGroup"/>
</xs:complexType>
</xs:element>
</xs:sequence>
<xs:attribute name="entityId" type="xs:anyURI"/>
</xs:complexType>
<xs:complexType name="nameType">
  <xs:sequence>
    <xs:annotation>
      <xs:documentation> name is used for a company name or if the given cannot be
        differentiated from the family name</xs:documentation>
    </xs:annotation>
    <xs:choice>
      <xs:element name="name" type="xs:string"/>
      <xs:sequence>
        <xs:element name="givenName" type="xs:string"/>
        <xs:element name="familyName" type="xs:string"/>
      </xs:sequence>
    </xs:choice>
  </xs:sequence>
  <xs:attribute name="id" type="xs:anyURI"/>
</xs:complexType>
<xs:complexType name="contactDetailsType">
  <xs:sequence>
    <xs:element name="name" type="nameType"/>
    <xs:element name="contactEmailAddress" type="xs:string" minOccurs="0">
      <xs:annotation>
        <xs:documentation>The e-mail address through which the contact can be directly
          accessed</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="address" type="addressType" minOccurs="0">
      <xs:annotation>
        <xs:documentation>The address of the contact. This is the personal address if
          provided. A professional address is provided in the organisation
          details</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="contactOccupation" type="xs:string" minOccurs="0">
      <xs:annotation>
        <xs:documentation>The job function of the contact</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="stageName" type="xs:string" minOccurs="0">
      <xs:annotation>
        <xs:documentation>For example, in the case the contact is a performing
          actor/actress, the stage name will be the fictitious character's
          name</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="relatedContacts" type="entityType"
      minOccurs="0" maxOccurs="unbounded">
      <xs:annotation>
        <xs:documentation> This is used to identify contacts related to the contact being
          described </xs:documentation>
      </xs:annotation>
    </xs:element>
  </xs:sequence>

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    <xs:attribute name="contactId" type="xs:anyURI"/>
  </xs:complexType>
  <xs:complexType name="organisationDetailsType">
    <xs:sequence>
      <xs:element name="organisationName" type="dc:elementType"/>
      <xs:element name="organisationDepartment" type="dc:elementType" minOccurs="0">
        <xs:annotation>
          <xs:documentation>To identify one or more production area / department / service
            where the resource was created/originated, in free text</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="organisationWebAddress" type="xs:string" minOccurs="0">
        <xs:annotation>
          <xs:documentation>The web address where additional information can be found
            regarding the company</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="address" type="addressType" minOccurs="0">
        <xs:annotation>
          <xs:documentation>The organisation address is also the professional address of the
            contact in the context of the content being described and in relation to the
            contact occupation provided in the contact details</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element name="contacts" type="entityType" minOccurs="0" maxOccurs="unbounded">
        <xs:annotation>
          <xs:documentation>Useful to provide contact information particularly is no other
            person information is otherwise provided.</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="organisationId" type="xs:anyURI"/>
  </xs:complexType>
  <xs:complexType name="addressType">
    <xs:annotation>
      <xs:documentation xml:lang="EN"> Provides address details for an organisation
    </xs:documentation>
    </xs:annotation>
    <xs:sequence>
      <xs:element name="addressLine" type="xs:string"
        minOccurs="0" maxOccurs="unbounded"/>
      <xs:element name="addressTownCity" type="xs:string" minOccurs="0"/>
      <xs:element name="addressCountyState" type="xs:string" minOccurs="0"/>
      <xs:element name="addressDeliveryCode" type="xs:string" minOccurs="0"/>
      <xs:element name="country" type="xs:string" minOccurs="0"/>
    </xs:sequence>
  </xs:complexType>
  <xs:attributeGroup name="typeGroup">
    <xs:attribute name="typeLabel" type="xs:string"/>
    <xs:attribute name="typeDefinition" type="xs:string"/>
    <xs:attribute name="typeLink" type="xs:anyURI"/>
  </xs:attributeGroup>
  <xs:attributeGroup name="formatGroup">
    <xs:attribute name="formatLabel" type="xs:string"/>
    <xs:attribute name="formatDefinition" type="xs:string"/>
    <xs:attribute name="formatLink" type="xs:anyURI"/>
  </xs:attributeGroup>
  <xs:attributeGroup name="statusGroup">
    <xs:attribute name="statusLabel" type="xs:string"/>
    <xs:attribute name="statusDefinition" type="xs:string"/>
  </xs:attributeGroup>

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```
<xs:attribute name="statusLink" type="xs:anyURI"/>
</xs:attributeGroup>
<xs:attributeGroup name="dateGroup">
  <xs:attribute name="startYear" type="xs:gYear"/>
  <xs:attribute name="startDate" type="xs:date"/>
  <xs:attribute name="startTime" type="xs:time"/>
  <xs:attribute name="endYear" type="xs:gYear"/>
  <xs:attribute name="endDate" type="xs:date"/>
  <xs:attribute name="endTime" type="xs:time"/>
  <xs:attribute name="period" type="xs:string"/>
</xs:attributeGroup>
</xs:schema>
```

